

Coordinates and Straight Lines

Choose the most appropriate option (a, b, c or d)

- Q 1. If two vertices of an equilateral triangle have integral coordinates then the third vertex will have
- (a) integral coordinates (b) coordinates which are rational
(c) at least one coordinate irrational (d) coordinates which are irrational
- Q 2. If the line segment joining (2, 3) and (-1, 2) is divided internally in the ratio 3 : 4 by the line $x + 2y = k$ then k is
- (a) $\frac{41}{7}$ (b) $\frac{5}{7}$ (c) $\frac{36}{7}$ (d) $\frac{31}{7}$
- Q 3. The polar coordinates of the vertices of a triangle are (0, 0), $(3, \pi/2)$ and $(3, \pi/6)$. Then the triangle is
- (a) right angled (b) isosceles (c) equilateral (d) none of these
- Q 4. The point (a, b + c), (b, c + a), (c, a + b) are
- (a) vertices of an equilateral triangle (b) collinear
(c) concyclic (d) none of these
- Q 5. The incentre of the triangle formed by the axes and the line $\frac{x}{a} + \frac{y}{b} = 1$ is
- (a) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (b) $\left(\frac{ab}{a+b+\sqrt{ab}}, \frac{ab}{a+b+\sqrt{ab}}\right)$
(c) $\left(\frac{a}{3}, \frac{b}{3}\right)$ (d) $\left(\frac{ab}{a+b+\sqrt{a^2+b^2}}, \frac{ab}{a+b+\sqrt{a^2+b^2}}\right)$
- Q 6. In the $\triangle ABC$, the coordinates of B are (0, 0), $AB = 2$, $\angle ABC = \frac{\pi}{3}$ and the middle point of BC has the coordinates (2, 0). The centroid of the triangle is
- (a) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ (b) $\left(\frac{5}{3}, \frac{1}{\sqrt{3}}\right)$ (c) $\left(\frac{4+\sqrt{3}}{3}, \frac{1}{3}\right)$ (d) none of these
- Q 7. The coordinates of three consecutive vertices of a parallelogram are (1, 3), (-1, 2) and (2, 5). The coordinates of the fourth vertex are
- (a) (6, 4) (b) (4, 6) (c) (-2, 0) (d) none of these
- Q 8. The area of the pentagon whose vertices are (4, 1), (3, 6), (-5, 1), (-3, -3) and (-3, 0) is
- (a) 30 unit² (b) 60 unit² (c) 120 unit² (d) none of these

- Q 9. A point moves in the x-y plane such that the sum of its distances from two mutually perpendicular lines is always equal to 3. The area enclosed by the locus of the point is
- (a) 18 unit² (b) $\frac{9}{2}$ unit² (c) 9 unit² (d) none of these
- Q 10. Let A = (1, 2), B = (3, 4) and let C = (x, y) be a point such that $(x - 1)(x - 3) + (y - 2)(y - 4) = 0$. If $\text{ar}(\triangle ABC) = 1$ then maximum number of positions of C in the x-y plane is
- (a) 2 (b) 4 (c) 8 (d) none of these
- Q 11. The points (α, β) , (γ, δ) , (α, δ) and (γ, β) taken in order, where $\alpha, \beta, \gamma, \delta$ are different real numbers, are
- (a) collinear (b) vertices of a square (c) vertices of a rhombus (d) concyclic
- Q 12. The diagonals of a parallelogram PQRS are along the lines $x + 3y = 4$ and $6x - 2y = 7$. The PQRS must be a
- (a) rectangle (b) square (c) cyclic quadrilateral (d) rhombus
- Q 13. The coordinates of the four vertices of a quadrilateral are (-2, 4), (-1, 2), (1, 2) and (2, 4) taken in order. The equation of the line passing through the vertex (-1, 2) and dividing the quadrilateral in two equal areas is
- (a) $x + 1 = 0$ (b) $x + y = 1$ (c) $x - y + 3 = 0$ (d) none of these
- Q 14. The equation of the straight line which passes through the point (-4, 3) such that the portion of the line between the axes is divided internally by the point in the ratio 5 : 3 is
- (a) $9x - 20y + 96 = 0$ (b) $9x + 20y = 24$ (c) $20x + 9y + 53 = 0$ (d) none of these
- Q 15. The equation of the straight line which bisects the intercepts made by the axes on the line $x + y = 2$ and $2x + 3y = 6$ is
- (a) $2x = 3$ (b) $y = 1$ (c) $2y = 3$ (d) $x = 1$
- Q 16. The equation of a straight line passing through the point (-2, 3) and making intercepts of equal length on the axes is
- (a) $2x + y + 1 = 0$ (b) $x - y = 5$ (c) $x - y + 5 = 0$ (d) none of these
- Q 17. The foot of the perpendicular on the line $3x + y = \lambda$ drawn from the origin is C. If the line cuts the x-axis and y-axis at A and B respectively then BC : CA is
- (a) 1 : 3 (b) 3 : 1 (c) 1 : 9 (d) 9 : 1
- Q 18. The distance of the line $2x - 3y = 4$ from the point (1, 1) in the direction of the line $x + y = 1$ is
- (a) $\sqrt{2}$ (b) $5\sqrt{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) none of these

- Q 19. The four sides of a quadrilateral are given by the equation $xy(x - 2)(y - 3) = 0$. The equation of the line parallel to $x - 4y = 0$ that divides the quadrilateral in two equal areas is
- (a) $x - 4y + 5 = 0$ (b) $x - 4y - 5 = 0$ (c) $4y = x + 1$ (d) $4y + 1 = x$
- Q 20. The coordinates of two consecutive vertices A and B of a regular hexagon ABCDEF are (1, 0) and (2, 0) respectively. The equation of the diagonal CE is
- (a) $\sqrt{3}x + y = 4$ (b) $x + \sqrt{3}y + 4 = 0$ (c) $x + \sqrt{3}y = 4$ (d) none of these
- Q 21. ABC is an isosceles triangle in which A is (-1, 0), $\angle A = 2\pi/3$, $AB = AC$ and AB is along the x-axis. If $BC = 4\sqrt{3}$ then the equation of the line BC is
- (a) $x + \sqrt{3}y = 3$ (b) $\sqrt{3}x + y = 3$ (c) $x + y = \sqrt{3}$ (d) none of these
- Q 22. The graph of the function $\cos x \cdot \cos(x + 2) - \cos^2(x + 1)$ is a
- (a) straight line passing through the point (0, $-\sin^2 1$) with slope 2
(b) straight line passing through the origin
(c) parabola with vertex (1, $-\sin^2 1$)
(d) straight line passing through the point ($\pi/2$, $-\sin^2 1$) and parallel to the x-axis
- Q 23. If the points (-2, 0), $(-1, 1/\sqrt{3})$ and $(\cos \theta, \sin \theta)$ are collinear then the number of value of $\theta \in [0, 2\pi]$ is
- (a) 0 (b) 1 (c) 2 (d) infinite
- Q 24. The limiting position of the point of intersection of the lines $3x + 4y = 1$ and $(1 + c)x + 3c^2y = 2$ as c tends to 1 is
- (a) (-5, 4) (b) (5, -4) (c) (4, -5) (d) none of these
- Q 25. The coordinate of the point on the x-axis which is equidistant from the points (-3, 4) and (2, 5) are
- (a) (20, 0) (b) (-23, 0) (c) $\left(\frac{4}{5}, 0\right)$ (d) none of these
- Q 26. The distance between the line $3x + 4y = 9$ and $6x + 8y + 15 = 0$ is
- (a) $\frac{3}{10}$ (b) $\frac{33}{10}$ (c) $\frac{33}{5}$ (d) none of these
- Q 27. If a vertex of an equilateral triangle is the origin and the side opposite to it has the equation $x + y = 1$ then the orthocenter of the triangle is
- (a) $\left(\frac{1}{3}, \frac{1}{3}\right)$ (b) $\left(\frac{\sqrt{2}}{3}, \frac{\sqrt{2}}{3}\right)$ (c) $\left(\frac{2}{3}, \frac{2}{3}\right)$ (d) none of these

- Q 28. The equation of the three sides of a triangle are $x = 2$, $y + 1 = 0$ and $x + 2y = 4$. The coordinates of the circumcentre of the triangle are
- (a) (4, 0) (b) (2, -1) (c) (0, 4) (d) none of these
- Q 29. L is a variable line such that the algebraic sum of the distances of the points (1, 1), (2, 0) and (0, 2) from the line is equal to zero. The line L will always pass through
- (a) (1, 1) (b) (2, 1) (c) (1, 2) (d) none of these
- Q 30. ABC is an equilateral triangle such that the vertices B and C lie on two parallel lines at a distance 6. If A lies between the parallel lines at a distance 4 from one of them the length of a side of the equilateral triangle is
- (a) 8 (b) $\sqrt{\frac{88}{3}}$ (c) $\frac{4\sqrt{7}}{\sqrt{3}}$ (d) none of these
- Q 31. If p and p' are the perpendiculars from the origin upon the lines $x \sec \theta + y \operatorname{cosec} \theta = a$ and $x \cos \theta - y \sin \theta = a \cos 2\theta$ respectively then
- (a) $4p^2 + p'^2 = a^2$ (b) $p^2 + 4p'^2 = a^2$ (c) $p^2 + p'^2 = a^2$ (d) none of these
- Q 32. Let the perpendiculars from any point on the line $2x + 11y = 5$ upon the lines $24x + 7y = 20$ and $4x - 3y = 2$ have the lengths p and p' respectively. Then
- (a) $2p = p'$ (b) $p = p'$ (c) $p = 2p'$ (d) none of these
- Q 33. If $P(1 + t/\sqrt{2})$ be any point on a line then the range of values of t for which the point P lies between the parallel lines $x + 2y = 1$ and $2x + 4y = 15$ is
- (a) $-\frac{4\sqrt{2}}{5} < t < \frac{5\sqrt{2}}{6}$ (b) $0 < t < \frac{5\sqrt{2}}{6}$ (c) $-\frac{4\sqrt{2}}{5} < t < 0$ (d) none of these
- Q 34. There are two parallel lines, one of which has the equation $3x + 4y = 2$. If the lines cut an intercept of length 5 on the line $x + y = 1$ then the equation of the other line is
- (a) $3x + 4y = \frac{\sqrt{6} - 2}{2}$ (b) $3x + 4y = \frac{\sqrt{6} + 2}{2}$ (c) $3x + 4y = 7$ (d) none of these
- Q 35. If the intercept made on the line $y = mx$ by lines $y = 2$ and $y = 6$ is less than 5 then the range of values of m is
- (a) $\left(-\infty, -\frac{4}{3}\right) \cup \left(\frac{4}{3}, +\infty\right)$ (b) $\left(-\frac{4}{3}, \frac{4}{3}\right)$ (c) $\left(-\frac{3}{4}, \frac{3}{4}\right)$ (d) none of these
- Q 36. If a, b, c are any three terms of an AP then the line $ax + by + c = 0$
- (a) has a fixed direction (b) always passes through a fixed point
- (c) always cuts intercepts on the axes such that their sum is zero

- (d) forms a triangle with the axes whose area is constant
- Q 37. If a, c, b are in GP then the line $ax + by + c = 0$
- (a) has a fixed direction (b) always passes through a fixed point
- (c) forms a triangle with the axes whose area is constant
- (d) always cuts intercepts on the axes such that their sum is zero
- Q 38. The number of real values of k for which the lines $x - 2y + 3 = 0$, $kx + 3y + 1 = 0$ and $4x - ky + 2 = 0$ are concurrent is
- (a) 0 (b) 1 (c) 2 (d) infinite
- Q 39. A family of lines is given by $(1 + 2\lambda)x + (1 - \lambda)y + \lambda = 0$, λ being the parameter. The line belonging to this family at the maximum distance from the point $(1, 4)$ is
- (a) $4x - y + 1 = 0$ (b) $33x + 12y + 7 = 0$ (c) $12x + 33y = 7$ (d) none of these
- Q 40. The members of the family of lines $(\lambda + \mu)x + (2\lambda + \mu)y = \lambda + 2\mu$, where $\lambda \neq 0$, $\mu \neq 0$, pass through the point
- (a) $(3, -1)$ (b) $(-3, 1)$ (c) $(1, 1)$ (d) none of these
- Q 41. The equations of the sides AB, BC and CA of the $\triangle ABC$ are $y - x = 2$, $x + 2y = 1$ and $3x + y + 5 = 0$ respectively. The equation of the altitude through B is
- (a) $x - 3y + 1 = 0$ (b) $x - 3y + 4 = 0$ (c) $3x - y + 2 = 0$ (d) none of these
- Q 42. The range of values of the ordinate of a point moving on the line $x = 1$, and always remaining in the interior of the triangle formed by the lines $y = x$, the x-axis and $x + y = 4$, is
- (a) $(0, 1)$ (b) $[0, 1]$ (c) $[0, 4]$ (d) none of these
- Q 43. If the point (a, a) falls between the lines $|x + y| = 2$ then
- (a) $|a| = 2$ (b) $|a| = 1$ (c) $|a| < 1$ (d) $|a| < \frac{1}{2}$
- Q 44. If $A(\sin \alpha, 1/\sqrt{2})$ and $B(1/\sqrt{2}, \cos \alpha)$, $-\pi \leq \alpha \leq \pi$, are two point on the same side of the line $x - y = 0$ then α belongs to the interval
- (a) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (b) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$ (c) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (d) none of these
- Q 45. The straight lines $L_1 \equiv 4x - 3y + 2 = 0$, $L_2 \equiv 3x + 4y - 4 = 0$ and $L_3 \equiv x - 7y + 6 = 0$
- (a) form a right-angled triangle (b) form a right-angled isosceles triangle
- (c) are concurrent (d) none of these

- Q 46. The equation of bisector of the acute angle between the lines $2x - y + 4 = 0$ and $x - 2y = 1$ is
 (a) $x + y + 5 = 0$ (b) $x - y + 1 = 0$ (c) $x - y = 5$ (d) none of these
- Q 47. The equation of the bisector of the acute angle between the lines $2x - y + 4 = 0$ and $x - 2y = 1$ is
 (a) $(\sqrt{5} - 2\sqrt{2})x + (\sqrt{5} + \sqrt{2})y = 3\sqrt{5} - 2\sqrt{2}$ (b) $(\sqrt{5} + 2\sqrt{2})x + (\sqrt{5} - \sqrt{2})y = 3\sqrt{5} + 2\sqrt{2}$
 (c) $3x = 10$ (d) none of these
- Q 48. Two lines $2x - 3y = 1$ and $x + 2y + 3 = 0$ divide the x-y plane in four compartments which are named as shown in figure. Consider the locations of the points $(2, -1)$, $(3, 2)$ and $(-1, -2)$. We get
 (a) $(2, -1) \in IV$ (b) $(3, 2) \in III$ (c) $(-1, -2) \in II$ (d) none of these
- Figure**
- Q 49. If the lines $y - x = 5$, $3x + 4y = 1$ and $y = mx + 3$ are concurrent then the value of m is
 (a) $\frac{19}{5}$ (b) 1 (c) $\frac{5}{19}$ (d) none of these
- Q 50. If the point $(\cos \theta, \sin \theta)$ does not fall in that angle between the lines $y = |x - 1|$ in which the origin lies then θ belongs to
 (a) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ (b) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (c) $(0, \pi)$ (d) none of these
- Q 51. The points $(-1, 1)$ and $(1, -1)$ are symmetrical about the line
 (a) $y + x = 0$ (b) $y = x$ (c) $x + y = 1$ (d) none of these
- Q 52. The equations of the line segment AB is $y = x$. If A and B lie on the same side of the line mirror $2x - y = 1$, the image of AB has the equation
 (a) $x + y = 2$ (b) $8x + y = 9$ (c) $7x - y = 6$ (d) none of these
- Q 53. Let $P = (1, 1)$ and $Q = (3, 2)$. The point R on the x-axis such that $PR + RQ$ is the minimum is
 (a) $\left(\frac{5}{3}, 0\right)$ (b) $\left(\frac{1}{3}, 0\right)$ (c) $(3, 0)$ (d) none of these
- Q 54. If a ray travelling along the $x = 1$ gets reflected from the line $x + y = 1$ then the equation of the line along which the reflected ray travels is
 (a) $y = 0$ (b) $x - y = 1$ (c) $x = 0$ (d) none of these
- Q 55. The point $P(2, 1)$ is shifted by $3\sqrt{2}$ parallel to the line $x + y = 1$, in the direction of increasing ordinate, to reach Q. The image of Q by the line $x = y = 1$ is

- (a) (5, -2) (b) (-1, 4) (c) (3, -4) (d) (-3, 2)

Q 56. Let A = (1, 0) and B = (2, 1). The line AB turns about A through an angle $\pi/6$ in the clockwise sense, and the new position of B is B'. Then B' has the coordinates

- (a) $\left(\frac{3+\sqrt{3}}{2}, \frac{\sqrt{3}-1}{2}\right)$ (b) $\left(\frac{3-\sqrt{3}}{2}, \frac{\sqrt{3}+1}{2}\right)$ (c) $\left(\frac{1-\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$ (d) none of these

Q 57. A line has intercepts a, b on the coordinate axes. If the axes are rotated about the origin through an angle α then the line has intercepts p, q on the new position of the axes respectively. Then

- (a) $\frac{1}{p^2} + \frac{1}{q^2} = \frac{1}{a^2} + \frac{1}{b^2}$ (b) $\frac{1}{p^2} - \frac{1}{q^2} = \frac{1}{a^2} - \frac{1}{b^2}$ (c) $\frac{1}{p^2} + \frac{1}{a^2} = \frac{1}{q^2} + \frac{1}{b^2}$ (d) none of these

Q 58. Two points A and B move on the x-axis and the y-axis respectively such that the distance between the two points is always the same. The locus of the middle point of AB is

- (a) a straight line (b) a pair of straight line (c) a circle (d) none of these

Q 59. Three vertices of a quadrilateral in order are (6, 1), (7, 2) and (-1, 0). If the area of the quadrilateral is 4 unit² then the locus of the fourth vertex has the equation

- (a) $x - 7y = 1$ (b) $x - 7y + 15 = 0$
(c) $(x - 7y)^2 + 14(x - 7y) - 15 = 0$ (d) none of these

Q 60. A variable line through the point (a, b) cuts the axes of reference at A and B respectively. The lines through A and B parallel to the y-axis and the x-axis respectively meet at P. Then the locus of P has the equation

- (a) $\frac{x}{a} + \frac{y}{b} = 1$ (b) $\frac{x}{b} + \frac{y}{a} = 1$ (c) $\frac{a}{x} + \frac{b}{y} = 1$ (d) $\frac{b}{x} + \frac{a}{y} = 1$

Choose the correct options. One or more options may be correct.

Q 61. If the coordinates of the vertices of a triangle are rational numbers then which of the following points of the triangle will always have rational coordinates ?

- (a) centroid (b) Incentre (c) Circumcentre (d) Orthocentre

Q 62. Two consecutive vertices of a rectangle of area 10 unit² are (1, 3) and (-2, -1). Other two vertices are

- (a) $\left(-\frac{3}{5}, \frac{21}{5}\right), \left(-\frac{18}{5}, \frac{1}{5}\right)$ (b) $\left(-\frac{3}{5}, -\frac{21}{5}\right), \left(-\frac{2}{5}, -\frac{11}{5}\right)$ (c) $\left(-\frac{2}{5}, -\frac{11}{5}\right), \left(\frac{13}{5}, \frac{9}{5}\right)$ (d) $\left(\frac{13}{5}, \frac{9}{5}\right), \left(-\frac{18}{5}, \frac{1}{5}\right)$

Q 63. The ends of a diagonal of a square are (2, -3) and (-1, 1). Another vertex of the square can be

- (a) $\left(-\frac{3}{2}, -\frac{5}{2}\right)$ (b) $\left(\frac{5}{2}, \frac{1}{2}\right)$ (c) $\left(\frac{1}{2}, \frac{5}{2}\right)$ (d) none of these

- Q 64. If each of the vertices of a triangle has integral coordinates then the triangle may be
 (a) right angled (b) equilateral (c) isosceles (d) none of these
- Q 65. If $(-1, 2)$, $(2, -1)$ and $(3, 1)$ are any three vertices of a parallelogram then the fourth vertex (a, b) will be such that
 (a) $a = 2, b = 0$ (b) $a = -2, b = 0$ (c) $a = -2, b = 6$ (d) $a = 6, b = -2$
- Q 66. If (α, β) be an end of a diagonal of a square and the other diagonal has the equation $x - y = \alpha$ then another vertex of the square can be
 (a) $(\alpha - \beta, \alpha)$ (b) $(\alpha, 0)$ (c) $(0, -\alpha)$ (d) $(\alpha + \beta, \beta)$
- Q 67. A point on the line $y = x$ whose perpendicular distance from the line $\frac{x}{4} + \frac{y}{3} = 1$ is 4 has the coordinates
 (a) $\left(-\frac{8}{7}, -\frac{8}{7}\right)$ (b) $\left(\frac{32}{7}, \frac{32}{7}\right)$ (c) $\left(\frac{3}{2}, \frac{3}{2}\right)$ (d) none of these
- Q 68. The parametric equation of a line is given by

$$x = -2 + \frac{r}{\sqrt{10}} \text{ and } y = 1 + 3\frac{r}{\sqrt{10}}$$

Then, for the line

- (a) intercept on the x-axis = $\frac{7}{3}$ (b) intercept on the y-axis = -7
 (c) slope of the line = $\tan^{-1} \frac{1}{3}$ (d) slope of the line = $\tan^{-1} 3$
- Q 69. One side of a square of length a is inclined to the x-axis at an angle α with one of the vertices of the square at the origin. The equation of a diagonal of the square is
 (a) $y(\cos \alpha - \sin \alpha) = x(\cos \alpha + \sin \alpha)$ (b) $y(\cos \alpha + \sin \alpha) = x(\cos \alpha - \sin \alpha)$
 (c) $y(\sin \alpha + \cos \alpha) - x(\sin \alpha - \cos \alpha) = a$ (d) $y(\sin \alpha + \cos \alpha) + x(\sin \alpha - \cos \alpha) = a$
- Q 70. If the equations of the hypotenuse and a side of a right-angled isosceles triangles be $x + my = 1$ and $x = k$ respectively then
 (a) $m = 1$ (b) $m = k$ (c) $m = -1$ (d) $m + k = 0$
- Q 71. The centroid and a vertex of an equilateral triangle are $(1, 1)$ and $(1, 2)$ respectively. Another vertex of the triangle can be
 (a) $\left(\frac{2 - \sqrt{3}}{2}, \frac{1}{2}\right)$ (b) $\left(\frac{2 + 3\sqrt{3}}{2}, \frac{1}{2}\right)$ (c) $\left(\frac{2 + \sqrt{3}}{2}, \frac{1}{2}\right)$ (d) none of these

- Q 72. If one vertex of an equilateral triangle of side 2 is the origin and another vertex lies on the line $x = \sqrt{3}y$ then the third vertex can be
- (a) (0, 2) (b) $(\sqrt{3}, -1)$ (c) (0, -2) (d) $(\sqrt{3}, 1)$
- Q 73. A line passing through the point (2, 2) and the axes enclose an area λ . The intercepts on the axes made by the line are given by the two roots of
- (a) $x^2 - 2|\lambda|x + |\lambda| = 0$ (b) $x^2 + |\lambda|x + 2|\lambda| = 0$ (c) $x^2 - |\lambda|x + 2|\lambda| = 0$ (d) none of these
- Q 74. A line passing through the origin and making an angle $\pi/4$ with the line $y - 3x = 5$ has the equation
- (a) $x + 2y = 0$ (b) $2x = y$ (c) $x = 2y$ (d) $y + 2x = 0$
- Q 75. The coordinates of a point on the line $x + y = 3$ such that the point is at equal distance from the line $|x| = |y|$ are
- (a) (3, 0) (b) (0, 3) (c) (-3, 0) (d) (0, -3)
- Q 76. A line perpendicular to the line $3x - 2y = 5$ cuts off an intercept 3 on the positive side of the x-axis. Then
- (a) the slope of the line is $\frac{2}{3}$ (b) the intercept on the y-axis is 2
- (c) the area of the triangle formed by the line with the axes is 3 unit² (d) none of these
- Q 77. One diagonal of a square is the portion of the line $\sqrt{3}x + y = 2\sqrt{3}$ intercepted by the axes. Then an extremity of the other diagonal is
- (a) $(1 + \sqrt{3}, \sqrt{3} - 1)$ (b) $(1 + \sqrt{3}, \sqrt{3} + 1)$ (c) $(1 - \sqrt{3}, \sqrt{3} - 1)$ (d) $(1 - \sqrt{3}, \sqrt{3} + 1)$
- Q 78. If $bx + cy = a$, where a, b, c are of the same sign, be a line such that the area enclosed by the line and the axes of reference is $\frac{1}{8}$ unit² then
- (a) b, a, c are in GP (b) a, 2a, c are in GP (c) b, $\frac{a}{2}$, c are in AP (d) b, -2a, c are in GP
- Q 79. The sides of a triangle are $x + y = 1$, $7y = x$ and $\sqrt{3}y + x = 0$. Then the following is an interior point of the triangle.
- (a) Circumcentre (b) Centroid (c) Incentre (d) Orthocentre
- Q 80. If (x, y) be a variable point on the line $y = 2x$ lying between the lines $2(x + 1) + y = 0$ and $x + 3(y - 1) = 0$ then
- (a) $x \in \left(-\frac{1}{2}, \frac{6}{7}\right)$ (b) $x \in \left(-\frac{1}{2}, \frac{3}{7}\right)$ (c) $y \in \left(-1, \frac{3}{7}\right)$ (d) $y \in \left(-1, \frac{6}{7}\right)$

Q 81. If the equations of the three sides of a triangle are $x + y = 1$, $3x + 5y = 2$ and $x - y = 0$ then the orthocenter of the triangle lies on the line

- (a) $5x - 3y = 2$ (b) $3x - 5y + 1 = 0$ (c) $2x - 3y = 1$ (d) $5x - 3y = 1$

Q 82. A ray travelling along the line $3x - 4y = 5$ after being reflected from a line l travels along the line $5x + 12y = 13$. Then the equation of the line l is

- (a) $x + 8y = 0$ (b) $x = 8y$ (c) $32x + 4y = 65$ (d) $32x - 4y + 65 = 0$

Q 83. A ray of light travelling along the line $x + y = 1$ is incident on the x -axis and after refraction it enters the other side of the x -axis by turning $\pi/6$ away from the x -axis. The equation of the line along which the refracted ray travels is

- (a) $x + (2 - \sqrt{3})y = 1$ (b) $(2 - \sqrt{3})x + y = 1$ (c) $y + (2 + \sqrt{3})x = 2 + \sqrt{3}$ (d) none of these

1c	2a	3c	4b	5d	6b	7b	8a	9a	10b
11d	12d	13c	14a	15b	16c	17d	18a	19a	20c
21a	22d	23b	24a	25d	26b	27a	28a	29a	30c
31a	32b	33a	34d	35a	36b	37c	38a	39c	40a
41b	42a	43c	44a	45c	46b	47a	48a	49c	50b
51b	52c	53a	54a	55d	56a	57a	58c	59c	60c
61acd	62ac	63ab	64acd	65bd	66bd	67ab	68d	69ac	70ac
71ac	72ab	73c	74cd	75ab	76bc	77bc	78bd	79bc	80bd
81bd	82bc	83ac							

Pair of Straight Lines

Choose the most appropriate option (a, b, c or d).

- Q 1. The equation $4x^2 + mxy - 3y^2 = 0$ represents a pair of real and distinct lines if
- (a) $m \in \mathbb{R}$ (b) $m \in (3, 4)$ (c) $m \in (-3, 4)$ (d) $m > 4$
- Q 2. If $2x^2 + 3xy + my^2 = 0$ represents two real and mutually perpendicular lines then m is
- (a) any negative real number (b) any positive real number
(c) -2 (d) none of these
- Q 3. The equation $kx^2 + 4xy + 5y^2 = 0$ represents two lines inclined at an angle π if k is
- (a) $\frac{5}{4}$ (b) $\frac{4}{5}$ (c) $-\frac{4}{5}$ (d) none of these
- Q 4. The equation $x^3 + y^3 = 0$ represents
- (a) three real straight lines (b) three points
(c) the combined equation of a straight line and a circle (d) none of these
- Q 5. The angle between the pair of lines $y^2 - 2xy \operatorname{cosec} \theta + x^2 = 0$, $0 \leq \theta \leq \frac{\pi}{2}$, is
- (a) $\frac{\pi}{2}$ (b) θ (c) $\frac{\pi}{2} - \theta$ (d) none of these
- Q 6. If the slope of one line is double the slope of another line and the combined equation of the pair of lines is $\frac{x^2}{a} + \frac{2xy}{h} + \frac{y^2}{b} = 0$ then $ab : h^2$ is
- (a) 9 : 8 (b) 3 : 2 (c) 8 : 3 (d) none of these
- Q 7. The triangle formed by the lines whose combined equation is $(y^2 - 4xy - x^2)(x + y - 1) = 0$ is
- (a) equilateral (b) right angled (c) isosceles (d) obtuse angled
- Q 8. The three lines whose combined equation is $y^3 - 4x^2y = 0$ form a triangle which is
- (a) isosceles (b) equilateral (c) right angled (d) none of these
- Q 9. The combined equation of the lines l_1, l_2 is $2x^2 + 6xy + y^2 = 0$ and that of the lines m_1, m_2 is $4x^2 + 18xy + y^2 = 0$. If the angle between l_1 and m_2 be α
- (a) $\frac{\pi}{2} - \alpha$ (b) 2α (c) $\frac{\pi}{4} + \alpha$ (d) α

- Q 10. The lines represented by $x^2 + 2\lambda xy + 2y^2 = 0$ and the lines represented by $(1 + \lambda)x^2 - 8xy + y^2 = 0$ are equally inclined then λ is
- (a) any real number (b) greater than 2 (c) ± 2 (d) less than -2
- Q 11. The area of the triangle formed by two rays whose combined equation is $y = |x|$ and the line $x + 2y = 2$ is
- (a) $\frac{8}{3} \text{ unit}^2$ (b) $\frac{4}{3} \text{ unit}^2$ (c) 4 unit^2 (d) $\frac{16}{3} \text{ unit}^2$
- Q 12. The centroid of the triangle whose three sides are given by the combined equation $(x^2 + 7xy + 2y^2)(y - 1) = 0$
- (a) $\left(\frac{2}{3}, 0\right)$ (b) $\left(\frac{7}{3}, \frac{2}{3}\right)$ (c) $\left(-\frac{7}{3}, \frac{2}{3}\right)$ (d) none of these
- Q 13. The orthocenter of the triangle formed by the pair of lines $2x^2 - xy - y^2 + x + 2y - 1 = 0$ and the line $x + y + 1 = 0$ is
- (a) (-1, 0) (b) (0, 1) (c) (-1, 1) (d) none of these
- Q 14. The angle between the pair of lines whose equation is $4x^2 + 10xy + my^2 + 5x + 10y = 0$ is
- (a) $\tan^{-1} \frac{3}{8}$ (b) $\tan^{-1} \frac{3}{4}$ (c) $\tan^{-1} \frac{2\sqrt{25-4m}}{m+4}$, $m \in \mathbb{R}$ (d) none of these
- Q 15. The combined equation of the pair of lines through the point (1, 0) and parallel to the lines represented by $2x^2 - xy - y^2 = 0$ is
- (a) $2x^2 - xy - 2y^2 + 4x - y = 6$ (b) $2x^2 - xy - y^2 - 4x - y + 2 = 0$
- (c) $2x^2 - xy - y^2 - 4x + y + 2 = 0$ (d) none of these
- Q 16. The equation $x^2 + (\lambda + \mu)xy + \lambda\mu y^2 + x + \mu y = 0$ represents two parallel straight lines if
- (a) $\lambda + \mu = 0$ (b) $\lambda = 4\mu$ (c) $\lambda = \mu$ (d) none of these
- Q 17. The product of perpendiculars drawn from the point (1, 2) to the pair of lines $x^2 + 4xy + y^2 = 0$ is
- (a) $\frac{9}{4}$ (b) $\frac{3}{4}$ (c) $\frac{9}{16}$ (d) none of these
- Q 18. If pair of lines represented by $ax^2 + 2hxy + by^2 = 0$, $b \neq 0$, are such that the sum of the slopes of the lines is three the product of their slopes then
- (a) $3b + 2h = 0$ (b) $3a + 2h = 0$ (c) $3h + 2a = 0$ (d) none of these
- Q 19. The pair of lines $\sqrt{3}x^2 - 4xy + \sqrt{3}y^2 = 0$ are rotated about the origin by $\frac{\pi}{6}$ in the anticlockwise sense. The equation of the pair in the new position is

- (a) $\sqrt{3}x^2 - xy = 0$ (b) $x^2 - \sqrt{3}xy = 0$ (c) $xy - \sqrt{3}y^2 = 0$ (d) none of these

Q 20. The equation of the image of the pair of rays $y = |x|$ by the line $x = 1$ is

- (a) $|y| = x + 2$ (b) $|y| + 2 = x$ (c) $y = |x - 2|$ (d) none of these

Q 21. Two lines represented by the equation $x^2 - y^2 - 2x + 1 = 0$ are rotated about the point $(1, 0)$, the line making the bigger angle with the positive direction of the x-axis being turned by 45° in the clockwise sense and the other line being turned by 15° in the anticlockwise sense. The combined equation of the pair of lines in their new positions is

- (a) $\sqrt{3}x^2 - xy + 2\sqrt{3}x - y + \sqrt{3} = 0$ (b) $\sqrt{3}x^2 - xy - 2\sqrt{3}x + y + \sqrt{3} = 0$
 (c) $\sqrt{3}x^2 - xy - 2\sqrt{3}x + \sqrt{3} = 0$ (d) none of these

Choose the correct options. One or more options may be correct.

Q 22. The equation $x^3 + x^2y - xy^2 = y^3$ represents

- (a) three real straight lines
 (b) lines in which two of them are perpendicular to each other
 (c) lines in which two of them are coincident
 (d) none of these

Q 23. The combined equation of two sides of an equilateral triangle is $x^2 - 3y^2 - 2x + 1 = 0$. If the length of a side of the triangle is 4 then the equation of the third side is

- (a) $x = 2\sqrt{3} + 1$ (b) $y = 2\sqrt{3} + 1$ (c) $x + 2\sqrt{3} = 1$ (d) $x = 2\sqrt{3}$

Q 24. Two pairs of straight lines have the equations $y^2 + xy - 12x^2 = 0$ and $ax^2 + 2hxy + by^2 = 0$. One line will be common among them if

- (a) $a = -3(2h + 3b)$ (b) $a = 8(h - 2b)$ (c) $a = 2(b + h)$ (d) $a = -3(b + h)$

Q 25. If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$ then m is

- (a) 1 (b) 2 (c) $-\frac{1}{2}$ (d) -1

Q 26. The straight lines represented by $x^2 + mxy - 2y + 3y - 1 = 0$ meet at

- (a) $\left(-\frac{1}{3}, \frac{2}{3}\right)$ (b) $\left(-\frac{1}{3}, -\frac{2}{3}\right)$ (c) $\left(\frac{1}{3}, \frac{2}{3}\right)$ (d) none of these

Q 27. If the chord $y = mx + 1$ of the circle $x^2 + y^2 = 1$ subtends an angle of measure 45° at the major segment of the circle then the value of m is

- (a) 2 (b) 1 (c) -1 (d) none of these

Q 28. The equation $2x^2 - 3xy - py^2 + x + qy - 1 = 0$ represents two mutually perpendicular lines if

- (a) $p = 3, q = 2$ (b) $p = 2, q = 3$ (c) $p = -2, q = 3$ (d) $p = 2, q = -\frac{9}{2}$

Q 29. The diagonals of a square are along the pair of lines whose equation is $2x^2 - 3xy - 2y^2 = 0$. If (2, 1) is a vertex of the square then another vertex consecutive to it can be

- (a) (1, -2) (b) (1, 4) (c) (-1, 2) (d) (-1, -4)

Q 30. There is a pair of points, one on each of the lines whose combined equation is $(4x - 3y + 5)(6x + 8y + 5) = 0$. If they are such that the distance of the point on one line is 2 units from the other line then the points are

- (a) $\left(\frac{1}{10}, \frac{9}{5}\right)$ and $\left(\frac{1}{2}, -1\right)$ (b) $\left(\frac{1}{2}, -1\right)$ and $\left(-\frac{23}{10}, -\frac{7}{3}\right)$ (c) $\left(\frac{1}{10}, \frac{9}{5}\right)$ and $\left(-\frac{23}{10}, -\frac{7}{5}\right)$ (d) none of these

Q 31. The pairs of straight lines $ax^2 + 2hxy - ay^2 = 0$ and $hx^2 - 2axy - hy^2 = 0$ are such that

- (a) one pair bisects the angles between the other pair
(b) the lines of one pair are equally inclined to the lines of the other pair
(c) the lines of one pair are perpendicular to the lines of the other pair
(d) none of these

Q 32. If the pair of lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ intersect on the y-axis then

- (a) $2fgh = bg^2 + ch^2$ (b) $bg^2 = ch^2$ (c) $abc = 2fgh$ (d) none of these

Q 33. The combined equation of three sides of a triangle is $(x^2 - y^2)(2x + 3y - 6) = 0$. If (-2, a) is an interior point and (b, 1) is an exterior point of the triangle then

- (a) $2 < a < \frac{10}{3}$ (b) $-2 < a < \frac{10}{3}$ (c) $-1 < b < \frac{9}{3}$ (d) $-1 < b < 1$

- 1a 2c 3b 4d 5c 6a 7b 8d 9d 10c
11b 12c 13a 14b 15c 16c 17a 18b 19a 20c
21b 22abc 23ac 24ab 25ad 26ac 27bc 28bd 29ac 30ab
31ab 32ab 33ad

Circles

Choose the appropriate option (a, b, c or d).

- Q 1. If the equation of a circle is $ax^2 + (2a - 3)y^2 - 4x - 1 = 0$ then its centre is
 (a) (2, 0) (b) (2/3, 0) (c) (-2/3, 0) (d) none of these
- Q 2. If $2x^2 + \lambda xy + 2y^2 + (\lambda - 4)x + 6y - 5 = 0$ is the equation of a circle then its radius is
 (a) $3\sqrt{2}$ (b) $2\sqrt{3}$ (c) $2\sqrt{2}$ (d) none of these
- Q 3. The equation $x^2 + y^2 - 2x + 4y + 5 = 0$ represents
 (a) a point (b) a pair of straight lines
 (c) a circle of nonzero radius (d) none of these
- Q 4. Three sides of a triangle have the equations $L_r \equiv y = m_r x - c_r = 0$; $r = 1, 2, 3$. Then $\lambda L_2 L_3 + \mu L_3 L_1 + \nu L_1 L_2 = 0$, where $\lambda \neq 0$, $\mu \neq 0$, $\nu \neq 0$, is the equation of the circumcircle of the triangle if
 (a) $\lambda(m_2 + m_3) + \mu(m_3 + m_1) + \nu(m_1 + m_2) = 0$ (b) $\lambda(m_2 m_3 - 1) + \mu(m_3 m_1 - 1) + \nu(m_1 m_2 - 1) = 0$
 (c) both (a) and (b) hold together (d) none of these
- Q 5. The number of integral values of λ for which

$$x^2 + y^2 + \lambda x + (1 - \lambda)y + 5 = 0$$
 is the equation of a circle whose radius cannot exceed 5, is
 (a) 14 (b) 18 (c) 16 (d) none of these
- Q 6. If $2(x^2 + y^2) + 4\lambda x + \lambda^2 = 0$ represents a circle of meaningful radius then the range of real values of λ is
 (a) R (b) $(0, +\infty)$ (c) $(-\infty, 0)$ (d) none of these
- Q 7. If a circle passes through the points of intersection of the lines $2x - y + 1 = 0$ and $x + \lambda y - 3 = 0$ with the axes of reference then the value of λ is
 (a) 1/2 (b) 2 (c) 1 (d) -2
- Q 8. The equation of the circle passing through the point (1, 1) and having two diameters along the pair lines $x^2 - y^2 - 2x + 4y - 3 = 0$ is
 (a) $x^2 + y^2 - 2x - 4y + 4 = 0$ (b) $x^2 + y^2 + 2x + 4y - 4 = 0$
 (c) $x^2 + y^2 - 2x + 4y + 4 = 0$ (d) none of these
- Q 9. Two vertices of an equilateral triangle are (-1, 0) and (1, 0), and its third vertex lies above the x-axis. The equation of the circumcircle of the triangle is

(a) $x^2 + y^2 = 1$

(b) $\sqrt{3}(x^2 + y^2) + 2y - \sqrt{3} = 0$

(c) $\sqrt{3}(x^2 + y^2) - 2y - \sqrt{3} = 0$

(d) none of these

Q 10. A triangle is formed by the lines whose combined equation is given by $(x + y - 4)(xy - 2x - y + 2) = 0$. The equation of its circumcircle is

(a) $x^2 + y^2 - 5x - 3y + 8 = 0$

(b) $x^2 + y^2 - 3x - 5y + 8 = 0$

(c) $x^2 + y^2 - 3x - 5y - 8 = 0$

(d) none of these

Q 11. If the centroid of an equilateral triangle is $(1, 1)$ and its one vertex is

(a) $x^2 + y^2 - 2x - 2y - 3 = 0$

(b) $x^2 + y^2 + 2x - 2y - 3 = 0$

(c) $x^2 + y^2 + 2x + 2y - 3 = 0$

(d) none of these

Q 12. The equation of the circle whose one diameter is PQ, where ordinates of P, Q are the roots of the equation $x^2 + 2x - 3 = 0$ and the abscissa are the roots of the equation $y^2 + 4y - 12 = 0$, is

(a) $x^2 + y^2 + 2x + 4y - 15 = 0$

(b) $x^2 + y^2 - 4x - 2y - 15 = 0$

(c) $x^2 + y^2 + 4x + 2y - 15 = 0$

(d) none of these

Q 13. The maximum number of points with rational coordinates on a circle whose centre is $(\sqrt{3}, 0)$ is

(a) one

(b) two

(c) four

(d) infinite

Q 14. A circle touches the y-axis at $(0, 2)$ and has an intercept of 4 units on the positive side of the x-axis. Then the equation of the circle is

(a) $x^2 + y^2 - 4(\sqrt{2}x + y) + 4 = 0$

(b) $x^2 + y^2 - 4(x + \sqrt{2}y) + 4 = 0$

(c) $x^2 + y^2 - 2(\sqrt{2}x + y)$

(d) none of these

Q 15. C_1 is a circle of radius 1 touching the x-axis and the y-axis. C_2 is another circle of radius > 1 and touching the axes as well as the circle C_1 . Then the radius of C_2 is

(a) $3 - 2\sqrt{2}$

(b) $3 + 2\sqrt{2}$

(c) $3 + 2\sqrt{3}$

(d) none of these

Q 16. The intercept on the line $y = x$ by the circle $x^2 + y^2 - 2x = 0$ is AB. The equation of the circle with AB as a diameter is

(a) $x^2 + y^2 + x + y = 0$

(b) $x^2 + y^2 = x + y$

(c) $x^2 + y^2 - 3x + y = 0$

(d) none of these

Q 17. Two circles, each of radius 5, have a common tangent at $(1, 1)$ whose equation is $3x + 4y - 7 = 0$. Then their centres are

(a) $(4, -5), (-2, 3)$

(b) $(4, -3), (-2, 5)$

(c) $(4, 5), (-2, -3)$

(d) none of these

- Q 18. The equation of the circumcircle of the regular hexagon whose two consecutive vertices have the coordinates $(-1, 0)$ and $(1, 0)$ and which lies wholly above the x-axis, is
- (a) $x^2 + y^2 - 2\sqrt{3}y - 1 = 0$ (b) $x^2 + y^2 - \sqrt{3}y - 1 = 0$
- (c) $x^2 + y^2 - 2\sqrt{3}x - 1 = 0$ (d) none of these
- Q 19. The equation of the incircle of the triangle formed by the axes and the line $4x + 3y = 6$ is
- (a) $x^2 + y^2 - 6x - 6y + 9 = 0$ (b) $4(x^2 + y^2 - x - y) + 1 = 0$
- (c) $4(x^2 + y^2 + x + y) + 1 = 0$ (d) none of these
- Q 20. IF p and q be the longest distance and the shortest distance respectively of the point $(-7, 2)$ from any point (α, β) on the curve whose equation is $x^2 + y^2 - 10x - 14y - 51 = 0$ then GM of p and q is equal to
- (a) $2\sqrt{11}$ (b) $5\sqrt{5}$ (c) 13 (d) none of these
- Q 21. The equation of the circumcircle of an equilateral triangle is $x^2 + y^2 + 2gx + 2fy + c = 0$ and one vertex of the triangle is $(1, 1)$. The equation of incircle of the triangle is
- (a) $4(x^2 + y^2) = g^2 + f^2$ (b) $4(x^2 + y^2) + 8gx + 8fy = (1 - g)(1 + 3g) + (1 - f)(1 + 3f)$
- (c) $4(x^2 + y^2) + 8gx + 8fy = g^2 + f^2$ (d) none of these
- Q 22. Let $f(x, y) = 0$ be the equation of a circle. If $(0, \lambda) = 0$ has equal roots $\lambda = 2, 2$ and $f(\lambda, 0) = 0$ has roots $\lambda = \frac{4}{5}, 5$ then the centre of the circle is
- (a) $(2, 29/10)$ (b) $(29/10, 2)$ (c) $(-2, 29/10)$ (d) none of these
- Q 23. For each $k \in \mathbb{N}$, let C_k denote the circle whose equation is $x^2 + y^2 = k^2$. On the circle C_k , a particle moves k units in the anticlockwise direction. After completing its motion on C_k , the particle moves to C_{k+1} in the radial direction. The motion of the particle continues in this manner. The particle starts at $(1, 0)$. If the particle crosses the positive direction of the x-axis for the first time on the circle C_n then n is
- (a) 7 (b) 6 (c) 2 (d) none of these
- Q 24. Two distinct chords drawn from the point (p, q) on the circle $x^2 + y^2 = px + qy$, where $pq \neq 0$, are bisected by the x-axis. Then
- (a) $|p| = |q|$ (b) $p^2 = 8q^2$ (c) $p^2 < 8q^2$ (d) $p^2 > 8q^2$
- Q 25. The length of the chord of the circle $x^2 + y^2 + 4x - 7y + 12 = 0$ along the y-axis is
- (a) 1 (b) 2 (c) $1/2$ (d) none of these

- Q 26. If the line $y - 1 = m(x - 1)$ cuts the circle $x^2 + y^2 = 4$ at two real points then the number of possible values of m is
- (a) 1 (b) 2 (c) infinite (d) none of these
- Q 27. The length of the chord of the circle $x^2 + y^2 = 9$ passing through $(3, 0)$ and perpendicular to the line $y + x = 0$ is
- (a) $3/\sqrt{2}$ (b) $3\sqrt{2}$ (c) $2\sqrt{3}$ (d) none of these
- Q 28. The number of points on the circle $2x^2 + 2y^2 - 3x = 0$ which are at a distance 2 from the point $(-2, 1)$ is
- (a) 2 (b) 0 (c) 1 (d) none of these
- Q 29. The equation of the diameter of the circle $3(x^2 + y^2) - 2x + 6y - 9 = 0$ which is perpendicular to the line $2x + 3y = 12$ is
- (a) $3x - 2y = 3$ (b) $3x - 2y + 1 = 0$ (c) $3x - 2y = 0$ (d) none of these
- Q 30. The equation of a circle C is $x^2 + y^2 - 6x - 8y - 11 = 0$. The number of real points at which the circle drawn with the points $(1, 8)$ and $(0, 0)$ at the ends of a diameter cuts the circle C is
- (a) 0 (b) 1 (c) 2 (d) none of these
- Q 31. The equation of the circle of radius $2\sqrt{2}$ whose centre lies on the line $x - y = 0$ and which touches the line $x + y = 4$, and whose centre's coordinates satisfy the inequality $x + y > 4$ is
- (a) $x^2 + y^2 - 8x - 8y + 24 = 0$ (b) $x^2 + y^2 = 8$
(c) $x^2 + y^2 - 8x + 8y = 24$ (d) none of these
- Q 32. The equation of the chord of the circle $x^2 + y^2 = 25$ of length 8 that passes through the point $(2\sqrt{3}, 2)$ and makes an acute angle with the positive direction of the x -axis is
- (a) $(4\sqrt{3} - 3\sqrt{7})x + 3y = 18 - 6\sqrt{21}$ (b) $(4\sqrt{3} + 3\sqrt{7})x - 3y = 18 + 6\sqrt{21}$
(c) $(4\sqrt{3} + 3\sqrt{7})x - 3y + 18 + 6\sqrt{21} = 0$ (d) none of these
- Q 33. If (a, b) is a point on the chord AB of the circle, where the ends of the chord are $A = (2, -3)$ and $B = (3, 2)$, then
- (a) $a \in [-3, 2], b \in [2, 3]$ (b) $a \in [2, 3], b \in [-3, 2]$
(c) $a \in [-2, 2], b \in [-3, 3]$ (d) none of these
- Q 34. The number of points with integral coordinates that are interior to the circle $x^2 + y^2 = 16$ is
- (a) 43 (b) 49 (c) 45 (d) 51

- Q 35. The range of values of a for which the point $(a, 4)$ is outside the circles $x^2 + y^2 + 10x = 0$ and $x^2 + y^2 - 12x + 20 = 0$ is
- (a) $(-\infty, -8) \cup (-2, 6) \cup (6, +\infty)$ (b) $(-8, -2)$
 (c) $(-\infty, -8) \cup (-2, +\infty)$ (d) none of these
- Q 36. A region in the x - y plane is bounded by the curve $y = \sqrt{25 - x^2}$ and the line $y = 0$. If the point $(a, a + 1)$ lies in the interior of the region then
- (a) $a \in (-4, 3)$ (b) $a \in (-\infty, -1) \cup (3, +\infty)$ (c) $a \in (-1, 3)$ (d) none of these
- Q 37. If $(2, 4)$ is a point interior to the circle $x^2 + y^2 - 6x - 10y + \lambda = 0$ and the circle does not cut the axes at any point then λ belongs to the interval
- (a) $(25, 32)$ (b) $(9, 32)$ (c) $(32, +\infty)$ (d) none of these
- Q 38. The range of values of $\theta \in [0, 2\pi]$ for which $(1 + \cos \theta, \sin \theta)$ is an interior point of the circle $x^2 + y^2 = 1$ is
- (a) $(\pi/6, 5\pi/6)$ (b) $(2\pi/3, 5\pi/3)$ (c) $(\pi/6, 7\pi/6)$ (d) $(2\pi/3, 4\pi/3)$
- Q 39. The range of the values of r for which the point $\left(-5 + \frac{r}{\sqrt{2}}, -3 + \frac{r}{\sqrt{2}}\right)$ is an interior point of the major segment of the circle $x^2 + y^2 = 16$, cut off by the line $x + y = 2$, is
- (a) $(-\infty, 5\sqrt{2})$ (b) $(4\sqrt{2} - \sqrt{14}, 5\sqrt{2})$ (c) $(4\sqrt{2} - \sqrt{14}, 4\sqrt{2} + \sqrt{14})$ (d) none of these
- Q 40. There are two circles whose equations are $x^2 + y^2 = 9$ and $x^2 + y^2 - 8x - 6y + n^2 = 0$, $n \in \mathbb{Z}$. If the two circles have exactly two common tangents then the number of possible values of n is
- (a) 2 (b) 8 (c) 9 (d) none of these
- Q 41. The number of common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is
- (a) 0 (b) 1 (c) 3 (d) 4
- Q 42. The number of common tangents to the circles $x^2 + y^2 + 2x + 8y - 23 = 0$ and $x^2 + y^2 - 4x - 10y + 19 = 0$ is
- (a) 1 (b) 2 (c) 3 (d) 4
- Q 43. If the circles $x^2 + y^2 + 2ax + c = 0$ and $x^2 + y^2 + 2by + c = 0$ touch each other then
- (a) $a^{-2} + b^{-2} = c^{-1}$ (b) $a^{-2} + b^{-2} = c^{-2}$ (c) $a + b = 2c$ (d) $\frac{1}{a} + \frac{1}{b} = \frac{2}{c}$
- Q 44. The number of common tangents to the circles one of the which passes through the origin and cuts off intercepts 2 from each of the axes, and the other circle has the line segment the origin and the point $(1, 1)$ as a diameter, is

- (a) 0 (b) 1 (c) 3 (d) 2

Q 45. The range of values of λ for which the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 4\lambda x + 9 = 0$ have two common tangents, is

- (a) $\in \left[-\frac{13}{8}, \frac{13}{8} \right]$ (b) $\lambda > \frac{13}{8}$ or $\lambda < -\frac{13}{8}$ (c) $1 < \lambda < \frac{13}{8}$ (d) none of these

Q 46. The number of common tangents to the circles $x^2 + y^2 - 6x - 14y + 48 = 0$ and $x^2 + y^2 - 6x = 0$ is

- (a) 1 (b) 2 (c) 0 (d) 4

Q 47. Two circles have the equations $x^2 + y^2 - 4x - 6y - 8 = 0$ and $x^2 + y^2 - 2x - 3 = 0$. Then

- (a) they cut each other (b) they touch each other
(c) one circle lies inside the other (d) one circle lies wholly outside the other

Q 48. The equations of two circles are $x^2 + y^2 - 26y + 25 = 0$ and $x^2 + y^2 = 25$. Then

- (a) they touches each other (b) they cut each other orthogonally
(c) one circle is inside the other circle (d) none of these

Q 49. A tangent is drawn to the circle $2(x^2 + y^2) - 3x + 4y = 0$ and it touches the circle at point A. The tangent passes the point P(2, 1). Then PA is equal to

- (a) 4 (b) 2 (c) $2\sqrt{2}$ (d) none of these

Q 50. If the points A(1, 4) and B are symmetrical about the tangent to the circles $x^2 + y^2 - x + y = 0$ at the origin then coordinates of B are

- (a) (1, 2) (b) $(\sqrt{2}, 1)$ (c) (4, 1) (d) none of these

Q 51. The range of values of m for which the line $y = mx + 2$ cuts the circle $x^2 + y^2 - x + y = 0$ at the origin then coordinates of B are

- (a) (1, 2) (b) $(\sqrt{2}, 1)$ (c) (4, 1) (d) none of these

Q 52. The range of values of m for which the line $y = mx + 2$ cuts the circles $x^2 + y^2 = 1$ at distinct or coincident points is

- (a) $(-\infty, -\sqrt{3}] \cup [\sqrt{3}, +\infty)$ (b) $[-\sqrt{3}, \sqrt{3}]$
(c) $[\sqrt{3}, +\infty)$ (d) none of these

Q 53. The equation of any tangent to the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ is

- (a) $y = m(x - 1) + 3\sqrt{1 + m^2} - 2$ (b) $y = mx + 3\sqrt{1 + m^2}$

- (c) $y = mx + 3\sqrt{1+m^2} - 2$ (d) none of these
- Q 54. Two tangents to the circle $x^2 + y^2 = 4$ at the point A and B meet at P(-4, 0). The area of the quadrilateral PAOB, where O is the origin, is
- (a) 4 (b) $6\sqrt{2}$ (c) $4\sqrt{3}$ (d) none of these
- Q 55. The angle between the pair of tangents from the point $(1, 1/2)$ to the circle $x^2 + y^2 + 4x + 2y - 4 = 0$ is
- (a) $\cos^{-1} \frac{4}{5}$ (b) $\sin^{-1} \frac{4}{5}$ (c) $\sin^{-1} \frac{3}{5}$ (d) none of these
- Q 56. The chords of contact of the pair of tangents to the circle $x^2 + y^2 = 1$ drawn from any point on the line $2x + y = 4$ pass through the point
- (a) $(1/2, 1/4)$ (b) $(1/4, 1/2)$ (c) $(1, 1/2)$ (d) $(1/2, 1)$
- Q 57. A foot of the normal from the point $(4, 3)$ to a circle is $(2, 1)$ and a diameter of the circle has the equation $2x - y = 2$. Then the equation of the circle is
- (a) $x^2 + y^2 + 2x - 1 = 0$ (b) $x^2 + y^2 - 2x - 1 = 0$ (c) $x^2 + y^2 - 2y - 1 = 0$ (d) none of these
- Q 58. The line $\lambda x + \mu y = 1$ is a normal to the circle $2x^2 + 2y^2 - 5x + 6y - 1 = 0$ if
- (a) $5\lambda - 6\mu = 2$ (b) $4 + 5\mu = 6\lambda$ (c) $4 + 6\mu = 5\lambda$ (d) none of these
- Q 59. The number of feet of normals from the point $(7, -4)$ to the circle $x^2 + y^2 = 5$ is
- (a) 1 (b) 2 (c) 3 (d) 4
- Q 60. The equation of a chord of the circle $x^2 + y^2 - 4x = 0$ which is bisected at the point $(1, 1)$ is
- (a) $x + y = 2$ (b) $3x - y = 2$ (c) $x - 2y + 1 = 0$ (d) $x - y = 0$
- Q 61. Lines are drawn through the point P(-2, -3) to meet the circle $x^2 + y^2 - 2x - 10y + 1 = 0$. The length of the line segment PA, A being the point on the circle where the line meets the circle at coincident points, is
- (a) 16 (b) $4\sqrt{3}$ (c) 48 (d) none of these
- Q 62. The equations of two circles are $x^2 + y^2 + 2\lambda x + 5 = 0$ and $x^2 + y^2 + 2\lambda y + 5 = 0$. P is any point on the line $x - y = 0$. If PA and PB are then lengths of the tangents from P to the two circles and $PA = 3$ then PB is equal to
- (a) 1.5 (b) 6 (c) 3 (d) none of these
- Q 63. The common chord of the circle $x^2 + y^2 + 6x + 8y - 7 = 0$ and a circle passing through the origin, the touching the line $y = x$, always passes through the point
- (a) $(-1/2, 1/2)$ (b) $(1, 1)$ (c) $(1/2, 1/2)$ (d) none of these

- Q 64. A tangent to the circle $x^2 + y^2 = 1$ through the point $(0, 5)$ cuts the circle $x^2 + y^2 = 4$ at A and B. The tangents to the circle $x^2 + y^2 = 4$ at A and B meet at C. The coordinates of C are
- (a) $\left(\frac{8}{5}\sqrt{6}, \frac{4}{5}\right)$ (b) $\left(\frac{8}{5}\sqrt{6}, -\frac{4}{5}\right)$ (c) $\left(-\frac{8}{5}\sqrt{6}, -\frac{4}{5}\right)$ (d) none of these
- Q 65. If the common chord of the circles $x^2 + (y - \lambda)^2 = 16$ and $x^2 + y^2 = 16$ subtend a right angle at the origin then λ is equal to
- (a) 4 (b) $4\sqrt{2}$ (c) $\pm 4\sqrt{2}$ (d) 8
- Q 66. The equation of the smallest circle passing through the intersection of the line $x + y = 1$ and the circle $x^2 + y^2 = 9$ is
- (a) $x^2 + y^2 + x + y - 8 = 0$ (b) $x^2 + y^2 - x - y - 8 = 0$ (c) $x^2 + y^2 - x + y - 8 = 0$
(d) none of these
- Q 67. The equation of a circle is $x^2 + y^2 = 4$. The centre of the smallest circle touching this circle and the line $x + y = 5\sqrt{2}$ has the coordinates
- (a) $\left(\frac{7}{2\sqrt{2}}, \frac{7}{2\sqrt{2}}\right)$ (b) $\left(\frac{3}{2}, \frac{3}{2}\right)$ (c) $\left(-\frac{7}{2\sqrt{2}}, -\frac{7}{2\sqrt{2}}\right)$ (d) none of these
- Q 68. The members of a family of circles are given by the equation $2(x^2 + y^2) + \lambda x - (1 + \lambda^2)y - 10 = 0$. The number of circles belonging to the family that are cut orthogonally by the fixed circle $x^2 + y^2 + 4x + 6y + 3 = 0$ is
- (a) 2 (b) 1 (c) 0 (d) none of these
- Q 69. The equation of the circle with the chord $y = 2x$ of the circle $x^2 + y^2 - 10x = 0$ as its diameter is
- (a) $x^2 + y^2 - 2x - 4y - 5 = 0$ (b) $x^2 + y^2 = 2x + 4y$
(c) $x^2 + y^2 = 4x + 2y$ (d) none of these
- Q 70. A circle of radius 2 touches the coordinate axes in the first quadrant. If the circle makes a complete rotation on the x-axis along the positive direction of the x-axis then the equation of the circle in the new position is
- (a) $x^2 + y^2 - 4(x + y) - 8\pi x + (2 + 4\pi)^2 = 0$ (b) $x^2 + y^2 - 4x - 4y + (2 + 4\pi)^2 = 0$
(c) $x^2 + y^2 - 8\pi x - 4y + (2 + 4\pi)^2 = 0$ (d) none of these
- Q 71. A ray of light incident at the point $(-2, -1)$ gets reflected from the tangent at $(0, -1)$ to the circle $x^2 + y^2 = 1$. The reflected ray touches the circle. The equation of the line along which the incident ray moved is
- (a) $4x - 3y + 11 = 0$ (b) $4x + 3y + 11 = 0$ (c) $3x + 4y + 11 = 0$ (d) none of these
- Q 72. The locus of the centres of the circles for which one end of a diameter is $(1, 1)$ while the other end is on the line $x + y = 3$ is

- (a) $x + y = 1$ (b) $2(x - y) = 5$ (c) $2x + 2y = 5$ (d) none of these
- Q 73. The angle between a pair of tangents drawn from a point P to the curve $x^2 + y^2 + 4x - 6y + 9\sin^2 \alpha + 13\cos^2 \alpha = 0$ is 2α . The locus of P is
- (a) $x^2 + y^2 + 4x - 6y + 4 = 0$ (b) $x^2 + y^2 + 4x - 6y - 9 = 0$
(c) $x^2 + y^2 + 4x - 6y - 4 = 0$ (d) $x^2 + y^2 + 4x - 6y + 9 = 0$
- Q 74. The point P moves in the plane of a regular hexagon such that the sum of the squares of its distances from the vertices of the hexagon is $6a^2$. If the radius of the circumcircles of the hexagon is $r (< a)$ then the locus of P is
- (a) a pair of straight lines (b) an ellipse
(c) a circle of radius $\sqrt{a^2 - r^2}$ (d) an ellipse of major axis a and minor axis r
- Q 75. The locus of the middle points of chords of length 4 of the circle $x^2 + y^2 = 16$ is
- (a) a straight line (b) a circle of radius 2 (c) a circle of radius $2\sqrt{3}$ (d) an ellipse
- Q 76. The equation of the locus of the middle point of a chord of the circle $x^2 + y^2 = 2(x + y)$ such that the pair of lines joining the origin to the point of intersection of the chord and the circle are equally inclined to the x-axis is
- (a) $x + y = 2$ (b) $x - y = 2$ (c) $2x - y = 1$ (d) none of these
- Q 77. The locus of the centres of circles passing through the origin and intersecting the fixed circle $x^2 + y^2 - 5x + 3y - 1 = 0$ orthogonally is
- (a) a straight line of the slope $\frac{3}{5}$ (b) a circle
(c) a pair of straight lines (d) none of these
- Q 78. The equation of a circle C_1 is $x^2 + y^2 - 4x - 2y - 11 = 0$. A circle C_2 of radius 1 rolls on the outside of the circle C_1 . The locus of the centre of C_2 has the equation
- (a) $x^2 + y^2 - 4x - 2y - 20 = 0$ (b) $x^2 + y^2 + 4x + 2y - 20 = 0$
(c) $x^2 + y^2 - 3x - y - 11 = 0$ (d) none of these
- Q 79. Circles are drawn through the point $(3, 0)$ to cut an intercept of length 6 units on the negative direction of the x-axis. The equation of the locus of their centres is
- (a) the x-axis (b) $x - y = 0$ (c) the y-axis (d) $x + y = 0$
- Q 80. The locus of the centre of a circle touching the line $x + 2y = 0$ and $x - 2y = 0$ is
- (a) $xy = 0$ (b) $x = 0$ (c) $y = 0$ (d) none of these

- Q 81. The locus of a point from which the lengths of the tangents to the circles $x^2 + y^2 = 4$ and $2(x^2 + y^2) - 10x + 3y - 2 = 0$ are equal to
- (a) a straight line inclined at $\pi/4$ with the line joining the centres of the circles
 (b) a circles (c) an ellipse
 (d) a straight line perpendicular to the line joining the centres of the circles
- Q 82. The locus of the centres of the circles passing through the intersection of the circles $x^2 + y^2 = 1$ and $x^2 + y^2 - 2x + y = 0$ is
- (a) a line whose equation is $x + 2y = 0$ (b) a line whose equation is $2x - y = 1$
 (c) a circle (d) a pair of lines

Choose the correct options. One or more options may be correct.

- Q 83. The equation of a circle is $x^2 + y^2 = 4$. A regular hexagon is inscribed in the circle whose one vertex is $(2, 0)$. Then a consecutive vertex has the coordinates
- (a) $(\sqrt{3}, 1)$ (b) $(1, -\sqrt{3})$ (c) $(\sqrt{3}, -1)$ (d) $(1, \sqrt{3})$
- Q 84. $P(\sqrt{2}, \sqrt{2})$ is a point on the circle $x^2 + y^2 = 4$ and Q is another point on the circle such that are $PQ = \frac{1}{4} \times \text{circumference}$. The coordinates of Q are
- (a) $(-\sqrt{2}, -\sqrt{2})$ (b) $(\sqrt{2}, -\sqrt{2})$ (c) $(-\sqrt{2}, \sqrt{2})$ (d) none of these
- Q 85. Circles $x^2 + y^2 = 1$ and $x^2 + y^2 - 8x + 11 = 0$ cut off equal intercepts on a line through the point $\left(-2, \frac{1}{2}\right)$. The slope of the line is
- (a) $\frac{-1 + \sqrt{29}}{14}$ (b) $\frac{1 + \sqrt{7}}{4}$ (c) $\frac{-1 - \sqrt{29}}{14}$ (d) none of these
- Q 86. Let L_1 be a straight line passing through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 and L_2 are equal then which of the following equations can represent L_1 ?
- (a) $x + y = 0$ (b) $x - y = 0$ (c) $x + 7y = 0$ (d) $x - 7y = 0$
- Q 87. The parametric equation of a circle is given by $x = 3 \cos \phi + 2$, $y = 3 \sin \phi$. Then is
- (a) centre = $(-2, 0)$ (b) radius = 3 (c) centre = $(2, 0)$ (d) radius = 1
- Q 88. If A and B are two points on the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ which are farthest and nearest respectively from the point $(7, 2)$ then

(a) $A = (2 - 2\sqrt{2}, -3 - 2\sqrt{2})$

(b) $B = (2 + 2\sqrt{2}, -3 + 2\sqrt{2})$

(c) $A = (2 + 2\sqrt{2}, -3 + 2\sqrt{2})$

(d) $B = (2 - 2\sqrt{2}, -3 + 2\sqrt{2})$

Q 89. A point $P(\sqrt{3}, 1)$ moves on the circle $x^2 + y^2 = 4$ and after covering a quarter of the circle leaves it tangentially. The equation of a line along which the point moves after leaving the circle is

(a) $y = \sqrt{3}x + 4$

(b) $\sqrt{3}y = x + 4$

(c) $\sqrt{3}y = x - 4$

(d) $y = \sqrt{3}x - 4$

Q 90. The equation of a circle of radius 1 touching the circles $x^2 + y^2 - 2|x| = 0$ is

(a) $x^2 + y^2 + 2\sqrt{3}x - 2 = 0$

(b) $x^2 + y^2 - 2\sqrt{3}y + 2 = 0$

(c) $x^2 + y^2 + 2\sqrt{3}y + 2 = 0$

(d) $x^2 + y^2 + 2\sqrt{3}x + 2 = 0$

Q 91. The line $4y - 3x + \lambda = 0$ touches the circle $x^2 + y^2 - 4x - 8y - 5 = 0$. The value of λ is

(a) 29

(b) 10

(c) -35

(d) none of these

Q 92. A circle which touches the axes, and whose centre is at distance $2\sqrt{2}$ from the origin, has the equation

(a) $x^2 + y^2 - 4x + 4y + 4 = 0$

(b) $x^2 + y^2 + 4x - 4y + 4 = 0$

(c) $x^2 + y^2 + 4x + 4y + 4 = 0$

(d) none of these

Q 93. Let the equation of a circle be $x^2 + y^2 = a^2$. If $h^2 + k^2 - a^2 < 0$ then the line $hx = ky = a^2$ is the

(a) polar line of the point (h, k) with respect to the circle

(b) real chord of contact of the tangents from (h, k) to the circle

(c) equation of a tangent to the circle from the point (h, k)

(d) none of these

Q 94. For the equation $x^2 + y^2 + 2\lambda x + 4 = 0$ which of the following can be true ?

(a) It represents a real circle for all $\lambda \in \mathbb{R}$.

(b) It represents a real circle for $|\lambda| > 2$

(c) The radical axis of any two circles of the family is the y-axis

(d) The radical axis of any two circles of the family is the x-axis.

Q 95. The point of contact of tangent from the point $(1, 2)$ to the circle $x^2 + y^2 = 1$ has the coordinates

(a) $\left(\frac{1-2\sqrt{19}}{5}, \frac{2+\sqrt{19}}{5}\right)$ (b) $\left(\frac{1-2\sqrt{19}}{5}, \frac{2-\sqrt{19}}{5}\right)$ (c) $\left(\frac{1+2\sqrt{19}}{5}, \frac{2+\sqrt{19}}{5}\right)$ (d) $\left(\frac{1+2\sqrt{19}}{5}, \frac{2-\sqrt{19}}{5}\right)$

Q 96. The equation of a circle C_1 is $x^2 + y^2 = 4$. The locus of the intersection of orthogonal tangents to the circles is the curve C_2 and the locus of the intersection of perpendicular tangents to the curve C_2 is the curve C_3 . Then

- (a) C_3 is a circle (b) the area enclosed by the curve C_3 is 8π
 (c) C_2 and C_3 are circles with the same centre (d) none of these

Q 97. A line parallel to the line $x - 3y = 2$ touches the circle $x^2 + y^2 - 4x + 2y - 5 = 0$ at the point

- (a) (1, -4) (b) (1, 2) (c) (3, -4) (d) (3, 2)

Q 98. A point on the line $x = 3$ from which the tangents drawn to the circle $x^2 + y^2 = 8$ are at right angles is

- (a) $(3, -\sqrt{7})$ (b) $(3, \sqrt{23})$ (c) $(3, \sqrt{7})$ (d) $(3, -\sqrt{23})$

Q 99. Tangents drawn from (2, 0) to the circle $x^2 + y^2 = 1$ touches the circle at A and B. Then

(a) $A = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), B = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ (b) $A = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right), B = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

(c) $A = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), B = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ (d) $A = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right), B = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

Q 100. The equations of four circles are $(x \pm a)^2 + (y \pm a)^2 = a^2$. The radius of a circle touching all the four circles is

- (a) $(\sqrt{2} - 1)a$ (b) $2\sqrt{2}a$ (c) $(\sqrt{2} + 1)a$ (d) $(2 + \sqrt{2})a$

Q 101. Let a line through the point P(5, 10) cut the line l whose equation is $x + 2y = 5$, at Q and the circle C whose equation is $x^2 + y^2 = 25$, at A and B. Then

- (a) P is the pole of the line l with respect to the circle C
 (b) l is the polar of the point P with respect to the circle C
 (c) PA, PQ, PB are in AP (d) PQ is the HM of PA and PB

Q 102. The equation of a circle is $x^2 + y^2 - 4x + 2y - 4 = 0$. With respect to the circle

- (a) the pole of the line $x - 2y + 5 = 0$ is (1, 1)
 (b) the chord of contact of real tangents from (1, 1) is the line $x - 2y + 5 = 0$
 (c) the polar of the point (1, 1) is $x - 2y + 5 = 0$
 (d) none of these

Answers

1b	2d	3a	4c	5c	6a	7d	8a	9c	10b
11a	12c	13b	14a	15b	16b	17c	18a	19b	20a
21b	22b	23a	24d	25a	26c	27b	28b	29a	30c
31a	32b	33b	34c	35a	36c	37a	38d	39b	40c
41b	42c	43a	44b	45b	46d	47a	48b	49b	50c
51a	52b	53a	54c	55b	56a	57b	58c	59b	60d
61b	62c	63c	64a	65c	66b	67a	68a	69b	70a
71b	72c	73d	74c	75c	76a	77d	78a	79c	80a
81d	82a	83bd	84bc	85ac	86bd	87bc	88ab	89bc	90bc
91ac	92bc	93a	94bc	95ad	96ac	97bc	98ac	99cd	100ac
101abd	102ac								

Parabola

Choose the most appropriate option (a, b, c or d).

- Q 1. Given the two ends of the latus rectum, the maximum number of parabolas that can be drawn is
(a) 1 (b) 2 (c) 0 (d) infinite
- Q 2. If the focus of a parabola is $(-2, 1)$ and the directrix has the equation $x + y = 3$ then the vertex is
(a) $(0, 3)$ (b) $(-1, 1/2)$ (c) $(-1, 2)$ (d) $(2, -1)$
- Q 3. If the vertex and the focus of a parabola are $(-1, 1)$ and $(2, 3)$ respectively then the equation of the directrix is
(a) $3x + 2y + 14 = 0$ (b) $3x + 2y - 25 = 0$ (c) $2x - 3y + 10 = 0$ (d) none of these
- Q 4. The vertex of a parabola is $(a, 0)$ and the directrix is $x + y = 3a$. The equation of the parabola is
(a) $x^2 + 2xy + y^2 + 6ax + 10ay + 7a^2 = 0$ (b) $x^2 - 2xy + y^2 + 6ax + 10ay + 2a^2 = 0$
(c) $x^2 - 2xy + y^2 - 6ax + 10ay = 2a^2$ (d) none of these
- Q 5. If the vertex = $(2, 0)$ and the extremities of the latus rectum are $(3, 2)$ and $(3, -2)$ then the equation of the parabola is
(a) $y^2 = 2x - 4$ (b) $x^2 = 4x - 8$ (c) $y^2 = 4x - 8$ (d) none of these
- Q 6. Any point on the parabola whose focus is $(0, 1)$ and the directrix is $x + 2 = 0$ is given by
(a) $(t^2 + 1, 2t - 1)$ (b) $(t^2 + 1, 2t + 1)$ (c) $(t^2, 2t)$ (d) $(t^2 - 1, 2t + 1)$
- Q 7. The equation of the parabola whose vertex and focus are on the positive side of the x-axis at distances a and b respectively from the origin is
(a) $y^2 = 4(b - a)(x - a)$ (b) $y^2 = 4(a - b)(x - b)$ (c) $x^2 = 4(b - a)(y - a)$ (d) none of these
- Q 8. The equation $x^2 + 4xy + 4y^2 - 3x - 6y - 4 = 0$ represents a
(a) circle (b) parabola (c) a pair of lines (d) none of these
- Q 9. The equation $\lambda x^2 + 4xy + y^2 + \lambda x + 3y + 2 = 0$ represents a parabola if λ is
(a) -4 (b) 4 (c) 0 (d) none of these
- Q 10. The focus of the parabola $y^2 - x - 2y + 2 = 0$ is
(a) $\left(\frac{5}{4}, 1\right)$ (b) $\left(\frac{1}{4}, 0\right)$ (c) $(1, 1)$ (d) none of these
- Q 11. The vertex of the parabola $(y - a)^2 = 4a(x + a)$ is

- (a) $(-a, a)$ (b) $(a, -a)$ (c) $(-2a, 2a)$ (d) $\left(-\frac{a}{2}, \frac{a}{2}\right)$

Q 12. The equation of the axis of the parabola $9y^2 - 16x - 12y - 57 = 0$ is

- (a) $2x = 3$ (b) $y = 3$ (c) $3y = 2$ (d) $x + 3y = 3$

Q 13. The length of the latus rectum of the parabola

$$169\{(x - 1)^2 + (y - 3)^2\} = (5x - 12y + 17)^2 \text{ is}$$

- (a) $\frac{14}{13}$ (b) $\frac{28}{13}$ (c) $\frac{12}{13}$ (d) none of these

Q 14. The length of the latus rectum of the parabola $x = ay^2 + by + c$ is

- (a) $\frac{a}{4}$ (b) $\frac{a}{3}$ (c) $\frac{1}{a}$ (d) $\frac{1}{4a}$

Q 15. The parametric equation of a parabola is $x = t^2 + 1$, $y = 2t + 1$. The Cartesian equation of its directrix is

- (a) $x = 0$ (b) $x + 1 = 0$ (c) $y = 0$ (d) none of these

Q 16. If $(2, -8)$ is at an end of a focal chord of the parabola $y^2 = 32x$ then the other end of the chord is

- (a) $(32, 32)$ (b) $(32, -32)$ (c) $(-2, 8)$ (d) none of these

Q 17. A line L passing through the focus of the parabola $y^2 = 4(x - 1)$ intersects the parabola in two distinct points. If 'm' be the slope of the line L then

- (a) $-1 < m < 1$ (b) $m < -1$ or $m > 1$ (c) $m \in \mathbb{R}$ (d) none of these

Q 18. The HM of the segments of a focal chord of the parabola $y^2 = 4ax$ is

- (a) $4a$ (b) $2a$ (c) a (d) a^2

Q 19. The length of a focal chord of the parabola $y^2 = 4ax$ at a distance b from the vertex is c . Then

- (a) $2a^2 = bc$ (b) $a^3 = b^2c$ (c) $ac = b^2$ (d) $b^2c = 4a^3$

Q 20. The parabola $y^2 = kx$ makes an intercept of length 4 on the line $x - 2y = 1$. Then k is

- (a) $\frac{\sqrt{105} - 5}{10}$ (b) $\frac{5 - \sqrt{105}}{10}$ (c) $\frac{5 + \sqrt{105}}{10}$ (d) none of these

Q 21. A double ordinate of the parabola $y^2 = 8px$ is of length 16p. The angle subtended by it at the vertex of the parabola is

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) none of these

- Q 22. The chord AB of the parabola $y^2 = 4ax$ cuts the axis of the parabola at C. If $A = (at_1^2, 2at_1)$, $B = (at_2^2, 2at_2)$ and $AC : AB = 1 : 3$ then
- (a) $t_2 = 2t_1$ (b) $t_2 + 2t_1 = 0$ (c) $t_1 + 2t_2 = 0$ (d) none of these
- Q 23. AB is a chord of the parabola $y^2 = 4ax$. If its equation is $y = mx + c$ and it subtends a right angle at the vertex of the parabola then
- (a) $c = 4am$ (b) $a = 4mc$ (c) $c = -4am$ (d) $a + 4mc = 0$
- Q 24. If ' t_1 ' and ' t_2 ' are the ends of a focal chord of the parabola $y^2 = 2x$ then
- (a) $t_1^2 + t_2^2 = 2$ (b) $t_1 + t_2 = 1$ (c) $t_1 t_2 = -1$ (d) none of these
- Q 25. A ray of light moving parallel to the x-axis gets reflected from a parabolic mirror whose equation is $(y - 2)^2 = 4(x + 1)$. After reflection, the ray must pass through the point
- (a) (0, 2) (b) (2, 0) (c) (0, -2) (d) (-1, 2)
- Q 26. The equation of a parabola is $y^2 = 4x$. P(1, 3) and Q(1, 1) are two points in the x-y plane. Then, for the parabola
- (a) P and Q are exterior points (b) P is an interior point while Q is an exterior point
- (c) P and Q are interior points (d) P is an exterior point while Q is an interior point
- Q 27. The point (a, 2a) is an interior point of the region bounded by the parabola $y^2 = 16x$ and the double ordinate through the focus. Then a belongs to the open interval
- (a) $a < 4$ (b) $0 < a < 4$ (c) $0 < a < 2$ (d) $a > 4$
- Q 28. The ends of a line segment are P(1, 3) and Q(1, 1). R is a point on the line segment PQ such that $PR : QR = 1 : \lambda$. If R is an interior point of the parabola $y^2 = 4x$ then
- (a) $\lambda \in (0, 1)$ (b) $\lambda \in \left(-\frac{3}{5}, 1\right)$ (c) $\lambda \in \left(\frac{1}{2}, \frac{3}{5}\right)$ (d) none of these
- Q 29. The range of values of λ for which the point $(\lambda, -1)$ is exterior to both the parabola $y^2 = |x|$ is
- (a) (0, 1) (b) (-1, 1) (c) (-1, 0) (d) none of these
- Q 30. The number of points with integral coordinates that lie in the interior of the region common to the circle $x^2 + y^2 = 16$ and the parabola $y^2 = 4x$ is
- (a) 8 (b) 10 (c) 16 (d) none of these
- Q 31. The number of distinct real tangents that can be drawn from (0, -2) to the parabola $y^2 = 4x$ is
- (a) one (b) two (c) zero (d) none of these
- Q 32. The tangent to the parabola $y^2 = 4x$ at the points (1, 2) and (4, 4) meet on the line

- (a) $x = 3$ (b) $x + y = 4$ (c) $y = 3$ (d) none of these
- Q 33. The point of intersection of the tangents to the parabola $y^2 = 4x$ at the points, where the parameter 't' has the value 1 and 2, is
 (a) (3, 8) (b) (1, 5) (c) (2, 3) (d) (4, 6)
- Q 34. The triangle formed by the tangents to a parabola $y^2 = 4ax$ at the ends of the latus rectum and the double ordinate through the focus is
 (a) equilateral (b) isosceles (c) right-angled isosceles
 (d) dependent on the value of a for its classification
- Q 35. If two tangents drawn from the point (α, β) to the parabola $y^2 = 4x$ be such that the slope of one tangent is double of the other then
 (a) $\beta = \frac{2}{9}\alpha^2$ (b) $\alpha = \frac{2}{9}\beta^2$ (c) $2\alpha = 9\beta^2$ (d) none of these
- Q 36. The tangents from the origin to the parabola $y^2 + 4 = 4x$ are inclined at
 (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$
- Q 37. If $y + b = m_1(x + a)$ and $y + b = m_2(x + a)$ are two tangent to the parabola $y^2 = 4ax$ then
 (a) $m_1 + m_2 = 0$ (b) $m_1 m_2 = 1$ (c) $m_1 m_2 = -1$ (d) none of these
- Q 38. The tangents to a parabola at the vertex V and any point P meet at Q. If S be the focus then SP, SQ, SV are in
 (a) AP (b) GP (c) HP (d) none of these
- Q 39. The equation of the common tangent to the equation parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is
 (a) $x + y + a = 0$ (b) $x + y = a$ (c) $x - y = a$ (d) none of these
- Q 40. 't₁' and 't₂' are two points on the parabola $y^2 = 4x$. If the chord joining them is a normal to the parabola at 't₁' then
 (a) $t_1 + t_2 = 0$ (b) $t_1(t_1 + t_2) = 1$ (c) $t_1(t_1 + t_2) + 2 = 0$ (d) $t_1 t_2 + 1 = 0$
- Q 41. The normal to the curve $x = at^2$, $y = 2at$ at the point P(t) meets the curve again at Q(t'). Then t' is
 (a) $t + \frac{1}{t}$ (b) $-t - \frac{2}{t}$ (c) $t + \frac{2}{t}$ (d) $t - \frac{1}{t}$
- Q 42. The set of points on the axis of the parabola $y^2 = 4x + 8$ from which the 3 normals to the parabola are all real and different is
 (a) $\{(k, 0) \mid k \leq -2\}$ (b) $\{(k, 0) \mid k > -2\}$ (c) $\{(0, k) \mid k > -2\}$ (d) none of these

- Q 43. The number of distinct normals that can be drawn from $(-2, 1)$ to the parabola $y^2 - 4x - 2y - 3 = 0$ is
- (a) 1 (b) 2 (c) 3 (d) 0
- Q 44. If the line $y = x + k$ is a normal to the parabola $y^2 = 4x$ then k can have the value
- (a) $2\sqrt{2}$ (b) 4 (c) -3 (d) 3
- Q 45. The arithmetic mean of the ordinates of the feet of the normals from $(3, 5)$ to the parabola $y^2 = 8x$ is
- (a) 4 (b) 0 (c) 8 (d) none of these
- Q 46. The area of the triangle formed by the tangent and the normal to the parabola $y^2 = 4ax$, both drawn at the same end of the latus rectum, and axis of the parabola is
- (a) $2\sqrt{2}a^2$ (b) $2a^2$ (c) $4a^2$ (d) none of these
- Q 47. If two of the three feet of normals drawn from a point to the parabola $y^2 = 4x$ be $(1, 2)$ and $(1, -2)$ then the third foot is
- (a) $(2, 2\sqrt{2})$ (b) $(2, -2\sqrt{2})$ (c) $(0, 0)$ (d) none of these
- Q 48. Let P, Q, R be three points on a parabola, normals at which are concurrent. The centroid of the ΔPQR must lie on
- (a) a line parallel to the directrix (b) the axis of the parabola
(c) a line of slope 1 passing through the vertex (d) none of these
- Q 49. The vertex of the parabola $y^2 = 8x$ is at the centre of a circle and the parabola cuts the circle at the ends of its latus rectum. Then the equation of the circle is
- (a) $x^2 + y^2 = 4$ (b) $x^2 + y^2 = 20$ (c) $x^2 + y^2 = 80$ (d) none of these
- Q 50. The length of the common chord of the parabola $2y^2 = 3(x + 1)$ and the circle $x^2 + y^2 + 2x = 0$ is
- (a) $\sqrt{3}$ (b) $2\sqrt{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) none of these
- Q 51. The circle $x^2 + y^2 + 2\lambda x = 0$, $\lambda \in \mathbb{R}$, touches the parabola $y^2 = 4x$ externally. Then
- (a) $\lambda > 0$ (b) $\lambda < 0$ (c) $\lambda > 1$ (d) none of these
- Q 52. The locus of the middle points of chords of a parabola which subtend a right angle at the vertex of the parabola is
- (a) a circle (b) an ellipse (c) a parabola (d) none of these
- Q 53. The locus of a point from which tangents to a parabola are at right angles is a

- (a) straight line (b) pair of straight line (c) circle (d) parabola
- Q 54. P is a point. Two tangents are drawn from it to the parabola $y^2 = 4x$ such that the slope of one tangent is three times the slope of the other. The locus of P is
- (a) a straight line (b) a circle (c) a parabola (d) an ellipse
- Q 55. The locus of the middle points of parallel chords of a parabola $x^2 = 4ay$ is a
- (a) straight line parallel to the x-axis (b) straight line parallel to the y-axis
- (c) circle 9d) straight line parallel to a bisector of the angles between the axes
- Q 56. The locus of the middle points of chords of the parabola $y^2 = 8x$ drawn through the vertex is a parabola whose
- (a) focus is (2, 0) (b) latus rectum = 8 (c) focus is (0, 2) (d) latus rectum = 4
- Q 57. The locus of the points of trisection of the double ordinates of a parabola is a
- (a) pair of lines (b) circle (c) parabola (d) straight line

Choose the correct options. One or more options may be correct.

- Q 58. The parabola $x^2 + 2x - 4y = 0$ has
- (a) vertex = (-1, -1) (b) latus rectum = 4 (c) focus = $\left(-1, \frac{3}{4}\right)$ (d) focus = $\left(0, -\frac{1}{4}\right)$
- Q 59. The equation of a parabola is $25\{(x - 2)^2 + (y + 5)^2\} = (3x + 4y - 1)^2$. For this parabola
- (a) vertex = (2, -5) (b) focus (2, -5) (c) directrix has the equation $3x + 4y - 1 = 0$
- (d) axis has the equation $3x + 4y - 1 = 0$
- Q 60. Let PQ be a chord of the parabola $y^2 = 4x$. A circle drawn with PQ as a diameter passes through the vertex V of the parabola. If ar (ΔPVQ) = 20 unit² then the coordinates of P are
- (a) (16, 8) (b) (16, -8) (c) (-16, 8) (d) (-16, -8)
- Q 61. The equation of a tangent to the parabola $y^2 = 9x$ from the point (4, 10) is
- (a) $x - 4y + 36 = 0$ (b) $81x - 8y - 162 = 0$ (c) $9x - 4y + 4 = 0$ (d) $x - 4y - 36 = 0$
- Q 62. If the tangents drawn from the point (0, 2) to the parabola $y^2 = 4ax$ are inclined at an angle $\frac{3\pi}{4}$ then the value of a is
- (a) 2 (b) -2 (c) 1 (d) none of these
- Q 63. If the tangents to the parabola $y^2 = 4ax$ at (x_1, y_1) , (x_2, y_2) cut at (x_3, y_3) then

(a) x_1, x_3, x_2 are in AP (b) x_1, x_3, x_2 are in GP (c) y_1, y_3, y_2 are in AP (d) y_1, y_3, y_2 are in GP

Q 64. The equation of a locus is $y^2 + 2ax + 2by + c = 0$. Then

(a) It is an ellipse (b) it is a parabola (c) its latus rectum = a (d) its latus rectum = $2a$

Q 65. A tangent to the parabola $y^2 = 4ax$ is inclined at $\frac{\pi}{3}$ with the axis of the parabola. The point of contact is

(a) $\left(\frac{a}{3}, -\frac{2a}{\sqrt{3}}\right)$ (b) $(3a, -2\sqrt{3}a)$ (c) $(3a, 2\sqrt{3}a)$ (d) $\left(\frac{3}{a}, \frac{2a}{\sqrt{3}}\right)$

Q 66. A chord PP' of a parabola cuts the axis of the parabola at O . The feet of the perpendicular from P and P' on the axis are M and M' respectively. If V is the vertex then VM, VO, VM' are in

(a) AP (b) GP (c) HP (d) none of these

Q 67. Let the equations of a circle and a parabola be $x^2 + y^2 - 4x - 6 = 0$ and $y^2 = 9x$ respectively. Then

(a) $(1, -1)$ is a point on the common chord of contact

(b) the equation of the common chord is $y + 1 = 0$

(c) the length of the common chord is 6

(d) none of these

Q 68. The equation of a common tangent to the parabola $y^2 = 2x$ and the circle $x^2 + y^2 + 4x = 0$ is

(a) $2\sqrt{6}x + y = 12$ (b) $x + 2\sqrt{6}y + 12 = 0$ (c) $x - 2\sqrt{6}y + 12 = 0$ (d) $2\sqrt{6}x - y = 12$

Q 69. Let there be two parabolas with the same axis, focus of each being exterior to the other and the latus recta being $4a$ and $4b$. The locus of the middle points of the intercepts between the parabolas made on the lines parallel to the common axis is a

(a) straight line if $a = b$ (b) parabola if $a \neq b$ (c) parabola for all a, b (d) none of these

Q 70. P is a point which moves in the x - y plane such that the point P is nearer to the centre of a square than any of the sides. The four vertices of the square are $(\pm a, \pm a)$. The region in which P will moved is bounded by parts of parabolas of which one has the equation

(a) $y^2 = a^2 + 2ax$ (b) $x^2 = a^2 + 2ay$ (c) $y^2 + 2ax = a^2$ (d) none of these

Answers

1b	2c	3a	4b	5c	6d	7a	8c	9b	10a
11a	12c	13b	14c	15a	16a	17d	18b	19d	20a
21b	22b	23c	24c	25a	26d	27b	28a	29b	30a
31b	32c	33c	34c	35b	36d	37c	38b	39a	40c
41b	42d	43a	44c	45b	46c	47c	48b	49b	50a
51a	52c	53a	54c	55b	56d	57c	58bc	59bc	60ab
61ac	62ab	63bc	64bd	65ad	66b	67ac	68bc	69ab	70abc

Ellipse and Hyperbola

Choose the most appropriate option (a, b, c or d)

- Q 1. The equation $2x^2 - 3xy + 5y^2 + 6x - 3y + 5 = 0$ represents
(a) a parabola (b) an ellipse (c) a hyperbola (d) a pair of straight lines
- Q 2. The set of real values of k for which the equation
 $(k + 1)x^2 + 2(k - 1)xy + y^2 - x + 2y + 3 = 0$
represents an ellipse is
(a) $(0, 3)$ (b) $(-\infty, 0)$ (c) $(3, +\infty)$ (d) $(-\infty, \infty)$
- Q 3. The centre of the conic section $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$ is
(a) $(2, 3)$ (b) $(2, -3)$ (c) $(-2, 3)$ (d) $(-2, -3)$
- Q 4. The eccentricity of the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ is
(a) $\frac{\sqrt{5}}{2}$ (b) $\frac{2}{3}$ (c) $\frac{\sqrt{5}}{3}$ (d) $\frac{4}{9}$
- Q 5. The eccentricity of the hyperbola $x^2 - 4y^2 = 16$ is
(a) 2 (b) $\frac{\sqrt{5}}{2}$ (c) 4 (d) $\frac{\sqrt{3}}{2}$
- Q 6. The eccentricity of the conic section $4(x^2 - y^2) = 1$ is
(a) $\sqrt{2}$ (b) 2 (c) 4 (d) $\frac{1}{4}$
- Q 7. The latus rectum of the conic section $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ whose eccentricity = 3, is
(a) $\frac{2a^2}{b}$ (b) $\frac{2b}{a^2}$ (c) $2a(1 - e^2)$ (d) $2b(1 - e^2)$
- Q 8. The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the point $(-3, 1)$ and has the eccentricity $\sqrt{\frac{2}{5}}$. Then the major axis of the ellipse has the length
(a) $4\sqrt{\frac{2}{5}}$ (b) $8\sqrt{\frac{2}{3}}$ (c) $4\sqrt{\frac{2}{3}}$ (d) $8\sqrt{\frac{2}{5}}$

- Q 9. The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through the point (2, 3) and has the eccentricity 2. Then the transverse axis of the hyperbola has the length
- (a) 1 (b) 3 (c) 2 (d) 4
- Q 10. In the ellipse $x^2 + 3y^2 = 9$ the distance between the foci is
- (a) $\sqrt{6}$ (b) 3 (c) $\frac{2}{3}\sqrt{6}$ (d) $2\sqrt{6}$
- Q 11. The minor axis of the ellipse $9x^2 + 5y^2 = 30$ is
- (a) 6 (b) $2\sqrt{5}$ (c) $\sqrt{6}$ (d) $\sqrt{5}$
- Q 12. The foci of the ellipse $25x^2 + 36y^2 = 225$ are
- (a) $\left(\pm\frac{1}{2}\sqrt{11}, 0\right)$ (b) $\left(\pm\frac{5}{2}, 0\right)$ (c) $\left(0, \pm\frac{1}{2}\sqrt{11}\right)$ (d) $\left(0, \pm\frac{5}{2}\right)$
- Q 13. If the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is e then the eccentricity of the hyperbola $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ is
- (a) e (b) $\frac{e}{\sqrt{e^2 - 1}}$ (c) $e\sqrt{e^2 - 1}$ (d) $e^2 - e$
- Q 14. If in an ellipse the minor axis = the distance between the foci and its latus rectum = 10 then the equation of the ellipse in the standard form is
- (a) $\frac{x^2}{(10)^2} + \frac{y^2}{(5\sqrt{2})^2} = 1$ (b) $\frac{x^2}{(5\sqrt{2})^2} + \frac{y^2}{(10)^2} = 1$ (c) $\frac{x^2}{25} + \frac{y^2}{(5/\sqrt{2})^2} = 1$ (d) none of these
- Q 15. If in a hyperbola the eccentricity is $\sqrt{3}$, and the distance between the foci is 9 then the equation of the hyperbola in the standard form is
- (a) $\frac{x^2}{\left(\frac{\sqrt{3}}{2}\right)^2} - \frac{y^2}{\left(\frac{\sqrt{3}}{2}\right)^2} = 1$ (b) $\frac{x^2}{\left(\frac{3\sqrt{3}}{2}\right)^2} - \frac{y^2}{\left(\frac{3\sqrt{3}}{\sqrt{2}}\right)^2} = 1$ (c) $\frac{x^2}{\left(\frac{3\sqrt{3}}{2}\right)^2} - \frac{y^2}{\left(\frac{3\sqrt{2}}{2}\right)^2} = 1$ (d) none of these
- Q 16. If in an ellipse, a focus is (6, 7), the corresponding directrix is $x + y + 2 = 0$ and the eccentricity = $\frac{1}{2}$ then the equation of the ellipse is
- (a) $7x^2 + 2xy + 7y^2 - 44x - 108y + 684 = 0$ (b) $7x^2 - 2xy + 7y^2 - 52x - 116y + 676 = 0$
- (c) $9x^2 - 2xy + 9y^2 - 44x - 108y + 684 = 0$ (d) none of these

- Q 17. If for a rectangular hyperbola a focus is $(1, 2)$ and the corresponding directrix is $x + y = 1$ then the equation of the rectangular hyperbola is
- (a) $x^2 - y^2 = 2$ (b) $xy - y + 2 = 0$ (c) $xy + y - 2 = 0$ (d) none of these
- Q 18. If two foci of an ellipse be $(-2, 0)$ and $(2, 0)$ and its eccentricity is $\frac{2}{3}$ then the ellipse has the equation
- (a) $5x^2 + 9y^2 = 45$ (b) $9x^2 = 5y^2 = 45$ (c) $5x^2 + 9y^2 = 90$ (d) $9x^2 + 5y^2 = 90$
- Q 19. If for a conic section a focus is $(-1, 1)$, eccentricity = 3 and the equation of the corresponding directrix is $x - y + 3 = 0$ then the equation of the conic section is
- (a) $7x^2 - 18xy + 7y^2 + 50x - 50y + 77 = 0$ (b) $7x^2 + 18xy + 7y^2 = 1$
(c) $7x^2 + 18xy + 7y^2 - 50x + 50y + 77 = 0$ (d) none of these
- Q 20. An ellipse having foci at $(3, 1)$ and $(1, 1)$ passes through the point $(1, 3)$. Its eccentricity is
- (a) $\sqrt{2} - 1$ (b) $\sqrt{3} - 1$ (c) $\frac{1}{2}(\sqrt{2} - 1)$ (d) $\frac{1}{2}(\sqrt{3} - 1)$
- Q 21. A point on the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$ at a distance 2 from the centre of the ellipse has the eccentric angle
- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$
- Q 22. A point P on the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ has the eccentric angle $\frac{\pi}{8}$. The sum of the distance of P from the two foci is
- (a) 5 (b) 6 (c) 10 (d) 3
- Q 23. If any point on a hyperbola has the coordinates $(5 \tan \phi, 4 \sec \phi)$ then the eccentricity of the hyperbola is
- (a) $\frac{5}{4}$ (b) $\frac{\sqrt{41}}{5}$ (c) $\frac{25}{16}$ (d) $\frac{\sqrt{41}}{4}$
- Q 24. The slope of the diameter of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose length is the GM of the major and minor axes, is
- (a) $\sqrt{\frac{a}{b}}$ (b) $\sqrt{ab}$ (c) $\sqrt{\frac{b}{a}}$ (d) $\frac{a}{b}$

- Q 25. PP' is a diameter of the ellipse $b^2x^2 + a^2y^2 = a^2b^2$ such that PP'^2 is the AM of the squares of the major and minor axes. Then the slope of PP' is
- (a) $\frac{b}{a}$ (b) $\frac{a}{b}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$
- Q 26. P is a variable point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$ whose foci are F_1 and F_2 . The maximum area (in unit)² of the $\triangle PFF'$ is
- (a) $2b\sqrt{a^2 - b^2}$ (b) $\sqrt{2}b\sqrt{a^2 - b^2}$ (c) $b\sqrt{a^2 - b^2}$ (d) $2a\sqrt{a^2 - b^2}$
- Q 27. Which of the following points is an exterior point of the ellipse $16x^2 + 9y^2 - 16x - 32 = 0$?
- (a) $\left(\frac{1}{2}, 2\right)$ (b) $\left(\frac{1}{4}, 1\right)$ (c) (3, -2) (d) none of these
- Q 28. For the hyperbola $\frac{x^2}{\cos^2 \alpha} - \frac{y^2}{\sin^2 \alpha} = 1$, which of the following remains constant when α varies ?
- (a) abscissa of vertices (b) abscissa of foci (c) eccentricity (d) directrix
- Q 29. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide. Then the value of b^2 is
- (a) 5 (b) 7 (c) 9 (d) 1
- Q 30. The equation of the tangent to the ellipse $4x^2 + 3y^2 = 12$ at the point whose eccentric angle is $\frac{\pi}{4}$ is
- (a) $\sqrt{3}x + 2y = 2\sqrt{6}$ (b) $2x + \sqrt{3}y = 2\sqrt{6}$ (c) $2x - \sqrt{3}y = 2\sqrt{6}$ (d) none of these
- Q 31. The number of values of m for which the line $y = mx + \sqrt{m^2 - 4}$ touches the hyperbola $4(x^2 - 1) = y^2$ is
- (a) two (b) zero (c) one (d) infinite
- Q 32. The value of c for which the line $y = 3x + c$ touches the ellipse $16x^2 + y^2 = 16$ is
- (a) 5 (b) 1 (c) 4 (d) 3
- Q 33. The number of values of $\phi \in [0, 2\pi]$ for which the line $2x \cos \phi + 3y \sin \phi = 6$ touches the ellipse $4x^2 + 9y^2 = 36$ is
- (a) four (b) two (c) one (d) infinite
- Q 34. The line $3x + 5y = k$ is a tangent to the ellipse $16x^2 + 25y^2 = 400$ if k is
- (a) $\pm\sqrt{5}$ (b) $\pm\sqrt{15}$ (c) ± 25 (d) $\pm\sqrt{10}$

- Q 35. The line $px + qy = r$ touches the hyperbola $b^2x^2 - a^2y^2 = a^2b^2$ if
 (a) $a^2p^2 + b^2q^2 = r^2$ (b) $a^2p^2 - b^2q^2 = r^2$ (c) $a^2q^2 + b^2p^2 = r^2$ (d) $a^2q^2 - b^2p^2 = r^2$
- Q 36. The equation of the tangent to the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, which is parallel to the line $y = 3x$, is
 (a) $y = 3x + \sqrt{241}$ (b) $y = 3x + 13$ (c) $y = 3x + \sqrt{209}$ (d) none of these
- Q 37. The equation of the tangent to the hyperbola $x^2 - 2y^2 = 18$, which is perpendicular to the line $x - y = 0$, is
 (a) $x + y = 3$ (b) $x + y + 2 = 0$ (c) $x + y = 3\sqrt{2}$ (d) $x + y + 3\sqrt{2} = 0$
- Q 38. If the tangents from the point $(\lambda, 3)$ to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ are at right angles then λ is
 (a) ± 1 (b) ± 3 (c) ± 2 (d) none of these
- Q 39. A point on the ellipse $x^2 + 3y^2 = 9$, where the tangent is parallel to the line $y - x = 0$, is
 (a) $(\sqrt{3}, \sqrt{2})$ (b) $\left(-\frac{3\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)$ (c) $\left(-\frac{3\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$ (d) $(-\sqrt{3}, \sqrt{2})$
- Q 40. The ordinate of the point of contact of a tangent is 2. Then the equation of the tangent to $x^2 + 4y^2 = 25$ is
 (a) $3x + 8y = 25$ (b) $8x + 3y = 25$ (c) $3x - 8y = 25$ (d) none of these
- Q 41. The tangent to the ellipse $16x^2 + 9y^2 = 144$, making equal intercepts on both the axes, is
 (a) $y = x + 3$ (b) $y = x - 2$ (c) $x + y = 5$ (d) $y = -x + 4$
- Q 42. If the tangent to the ellipse $x^2 + 4y^2 = 16$ at the point ' ϕ ' is a normal to the circle $x^2 + y^2 - 8x - 4y = 0$ then ϕ is equal to
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $-\frac{\pi}{4}$
- Q 43. The area of the quadrilateral formed by tangents at the end points of latus recta of the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ is
 (a) $\frac{27}{4}$ unit² (b) 9 unit² (c) $\frac{27}{2}$ unit² (d) 27 unit²
- Q 44. The tangent at $(3\sqrt{3} \cos \theta, \sin \theta)$ is drawn to the ellipse $\frac{x^2}{27} + y^2 = 1$. Then the value of θ such that the sum of intercepts on axes made by the tangent is minimum is

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{8}$ (d) $\frac{\pi}{4}$

Q 45. The number of normals that can be drawn to the curve $4x^2 + 9y^2 = 36$ from an external point, in general, is

- (a) 1 (b) 3 (c) 4 (d) infinite

Q 46. The number of distinct normal lines from the exterior point $(0, c)$, $c > b$, to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

- (a) 3 (b) 4 (c) 2 (d) 1

Q 47. The equation of the normal to the ellipse $x^2 + 4y^2 = 16$ at the end of the latus rectum in the first quadrant is

- (a) $2x + \sqrt{3}(y + 3) = 0$ (b) $2x = \sqrt{3}(y + 3)$ (c) $\sqrt{3}x = 2(y + 3)$ (d) none of these

Q 48. If the tangent and the normal to $x^2 - y^2 = 4$ at a point cut off intercepts a_1, a_2 on the x-axis respectively and b_1, b_2 on the y-axis respectively then the value of $a_1a_2 + b_1b_2$ is

- (a) 1 (b) -1 (c) 0 (d) 4

Q 49. The normal to the rectangular hyperbola $xy = c^2$ at the point ' t_1 ' meets the curve again at the point ' t_2 '. The value of $t_1^3 \cdot t_2$ is

- (a) 1 (b) c (c) $-c$ (d) -1

Q 50. If P and Q are the ends of a pair of conjugate diameter and C is the centre of the ellipse $4x^2 + 9y^2 = 36$ then the area of the ΔCPQ is

- (a) 6 unit² (b) 3 unit² (c) 2 unit² (d) 12 unit²

Q 51. If $y = x$ and $3y + 2x = 0$ are the equations of a pair of conjugate diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then its eccentricity is

- (a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{\sqrt{3}}{2}$

Q 52. The locus of a point $P(\alpha, \beta)$ moving under the condition that the line $y = \alpha x + \beta$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

- (a) a circle (b) an ellipse (c) a hyperbola (d) a parabola

Choose the correct option. One or more option may be correct.

Q 53. A focus of the hyperbola $25x^2 - 36y^2 = 225$ is

- (a) $(\sqrt{61}, 0)$ (b) $\left(\frac{1}{2}\sqrt{61}, 0\right)$ (c) $(-\sqrt{61}, 0)$ (d) $\left(-\frac{1}{2}\sqrt{61}, 0\right)$

Q 54. The point P on the ellipse $4x^2 + 9y^2 = 36$ is such that the area of the $\triangle PF_1F_2 = \sqrt{10}$ where F_1, F_2 are foci. Then P has the coordinates

- (a) $\left(\frac{3}{\sqrt{2}}, \sqrt{2}\right)$ (b) $\left(\frac{3}{2}, 2\right)$ (c) $\left(-\frac{3}{2}, -2\right)$ (d) $\left(-\frac{3}{\sqrt{2}}, -\sqrt{2}\right)$

Q 55. A point on the ellipse $x^2 + 3y^2 = 37$, where the normal is parallel to the line $6x - 5y = 2$, is

- (a) (5, -2) (b) (5, 2) (c) (-5, 2) (d) (-5, -2)

Answers

1b	2a	3a	4c	5b	6a	7c	8b	9c	10d
11b	12a	13b	14a	15b	16b	17c	18a	19a	20a
21a	22c	23d	24c	25a	26a	27c	28b	29b	30b
31d	32a	33d	34c	35b	36a	37a	38c	39c	40a
41c	42a	43d	44b	45c	46d	47b	48c	49d	50b
51c	52c	53bd	54ad	55bd					